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MaizeFormerX: a lightweight vision transformer with cross-scale attention for explainable maize leaf disease diagnosis

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Early detection of maize leaf diseases is essential to prevent yield losses. Existing vision-based models face challenges in real-world environments due to data imbalance, lighting variations, and interpretability. This study presents MaizeFormerX, a lightweight Vision Transformer designed for cross-domain, explainable maize disease detection on resource-limited settings. MaizeFormerX employs multi-scale patch embeddings and a Cross-Scale Attention Fusion (CSAF) module to capture both detailed lesion textures and larger disease patterns. The CSAF output is processed through a transformer encoder stack using multi-head self-attention to model long-range dependencies. Robust preprocessing and dataset-specific augmentations were applied to improve feature extraction and address class imbalances in the Dataverse, Tanzania, and Plagues Maiz datasets. For interpretability, Grad-CAM was used for pixel-level saliency mapping in an efficient web application. When benchmarked against MobileViT, EfficientFormer, TinyViT, and Swin Transformer, MaizeFormerX achieved 97.8% accuracy on Dataverse, 97.5% on Tanzania, and 96.9% on Plagues Maiz, outperforming Swin Transformer V2 by 2–3%. Cross-domain testing yielded 88.9% accuracy when trained on Dataverse and tested on Tanzania, surpassing baseline performance by 3–6%. Class-wise analysis revealed F1 scores over 98% for Healthy and MLB classes with 6× augmentation, and over 97% for MSV. Ablation studies highlighted the significance of the cross-scale attention module for high MCC during domain shifts. This study introduces a precise, explainable, and efficient image-based method for classifying maize diseases, which could aid in more targeted crop management, reduce unnecessary agrochemical use, and promote sustainable maize production in future decision-support environments.

Keywords maize leaf disease, vision transformer, cross-domain generalization, explainable AI, precision agriculture

In 2023, maize (Zea mays) production surpassed 1.2 billion tonnes, making it the most significant contributor to the record world cereal harvest of 2.8 billion tonnes[1,2]. This represents a substantial portion of the 61 million-tonne year-on-year increase in grain output[3]. Since 2010, maize production has expanded by 27% and consistently accounts for over 40% of all coarse grains grown worldwide[4,5]. Approximately two-thirds are used

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as feed for livestock and aquaculture, serving as the caloric foundation of global meat, dairy, and fish value chains[6,7]. Besides, starch and high-fructose syrup are commonly found in processed foods, pharmaceuticals, and biodegradable plastics, while the germ provides a valuable vegetable oil[8,9]. Importantly, modern bio-refineries in the United States, Brazil, China, and the EU convert more than 15% of the maize harvest into ethanol, positioning maize as a necessary feedstock for low-carbon transport fuels and improving rural energy security[10]. In 2024, the crop is projected to generate over US $220 billion in combined farm-gate revenue and trade receipts[11].

Unlike wheat or rice, maize is a thermophilic C4 plant whose leaf architecture has a large, photosynthetically active surface. Productivity is increasingly threatened by various biotic stresses that compromise both crop yield and quality[12]. Among these threats, Maize Lethal Necrosis (MLN), Maize Streak Virus (MSV), and Maize Leaf Blight (MLB) are particularly destructive, especially in tropical and subtropical regions[13]. These diseases, caused by viruses, fungi, and bacteria, target maize plants primarily through their photosynthetically active leaves[14]. While these leaves are essential for enhancing yield, they also render the plants more vulnerable to infections. The impact of these diseases can be severe, with outbreaks causing yield reductions between 30% and over 80%, depending on the severity of the epidemic and the susceptibility of the maize cultivar[15]. For example, MLN can devastate entire fields within weeks due to its synergistic viral infection, while MSV spreads quickly through insect vectors, resulting in stunted growth and chlorotic striping[16]. In many low- and middle-income countries, limited access to timely agronomic interventions exacerbates the threat of these diseases, directly impacting food security and farmers' livelihoods[17].

These diseases also diminish the quality of the grain, reducing its market value for industrial applications such as ethanol production or food processing. The visual similarities between different maize leaf diseases can often lead to misdiagnosis. This challenge is compounded by how these diseases can co-occur or develop under various environmental conditions, making visual differentiation even more complex. For example, the chlorosis associated with MSV can mimic the early signs of nutrient deficiency or herbicide injury. At the same time, lesions from MLB may initially be confused with physical damage or drought stress[18]. As a result, farmers and even trained extension officers may find it challenging to make accurate assessments in the field. The consequences of misidentification can be severe. Incorrect or delayed identification can allow diseases to spread unchecked, wasting valuable resources on ineffective treatments. In regions with limited access to plant pathologists, this may lead to the loss of planting material and have cascading effects on food and feed availability[19]. Therefore, detecting and classifying maize leaf diseases accurately and early is crucial. A robust system should be able to distinguish between visually similar symptoms across various conditions. There is a growing need to develop vision-based disease identification systems that can operate reliably in real-world agricultural settings.

Traditional methods are increasingly falling short in meeting the demands of large-scale agriculture[20]. These techniques are labor-intensive and highly subjective. The reliance on expert knowledge makes these methods non-scalable, especially in rural areas where trained personnel are scarce. Additionally, variations in disease manifestation due to environmental factors or different stages of plant growth further complicate the situation, reducing the reliability of visual assessments. Deep learning models have gained popularity for plant disease classification tasks to overcome these limitations. Convolutional Neural Networks (CNNs) have shown high accuracy by effectively extracting hierarchical spatial features from images[21]. However, CNNs inherently focus on local receptive fields, which restricts their ability to capture global contextual information, a crucial factor distinguishing visually similar or overlapping disease symptoms across large leaf surfaces. This can lead to misclassifications, particularly when understanding spatial relationships or subtle disease patterns requires a broader context.

By utilizing self-attention mechanisms, Vision Transformers (ViTs) effectively model long-range dependencies and capture global feature relationships, making them well-suited for fine-grained classification tasks. Despite their success in general computer vision benchmarks, ViTs remain underexplored in maize pathology. Moreover, most existing models are trained and validated exclusively on the PlantVillage dataset, which contains uniformly lit images with simple backgrounds. While this dataset is useful for initial benchmarking, it does not accurately reflect the noise, lighting variability, occlusion, and background complexity encountered in real-world environments. As a result, many models fail to generalize when applied outside their training domain. The lack of cross-dataset validation and reliable tools in current studies is equally concerning. Few models have been tested for domain adaptation, and even fewer incorporate explainability or usability features necessary for effective decision support in the field. This creates a significant gap between academic model performance and practical impact in crop disease management.

To tackle the challenges in maize pathology, we propose a novel transformer-based architecture called MaizeFormerX. This methodology (Fig. 1) is designed to function effectively under the complex and variable conditions associated with maize diseases. Unlike traditional CNNs, which focus primarily on local texture features, or monolithic ViTs that require extensive computational resources, MaizeFormerX is engineered to capture multi-scale lesion patterns while resolving contextual ambiguities across the leaf surface. The core innovation of MaizeFormerX lies in its integration of multi-scale patch embedding and cross-scale attention fusion. This allows the model to effectively represent fine-grained and global disease signatures with minimal parameter overhead.

The primary objective of this study is to ensure improved generalization across diverse agroecological domains. To achieve this, we rigorously evaluate the model using three distinct maize leaf datasets: Dataverse (lab-captured images), Tanzania (real-world field conditions), and Plagues-Maiz (fine-grained microbial classes). Our proposed MaizeFormerX model is a novel hybrid ViT architecture that combines multi-scale patch embedding with cross-scale attention fusion. This design allows the model to effectively capture both local lesion textures and the broader context of diseases. These innovations lead to strong performance in both clean and noisy environments, surpassing existing lightweight CNN and ViT baselines. In addition to architectural

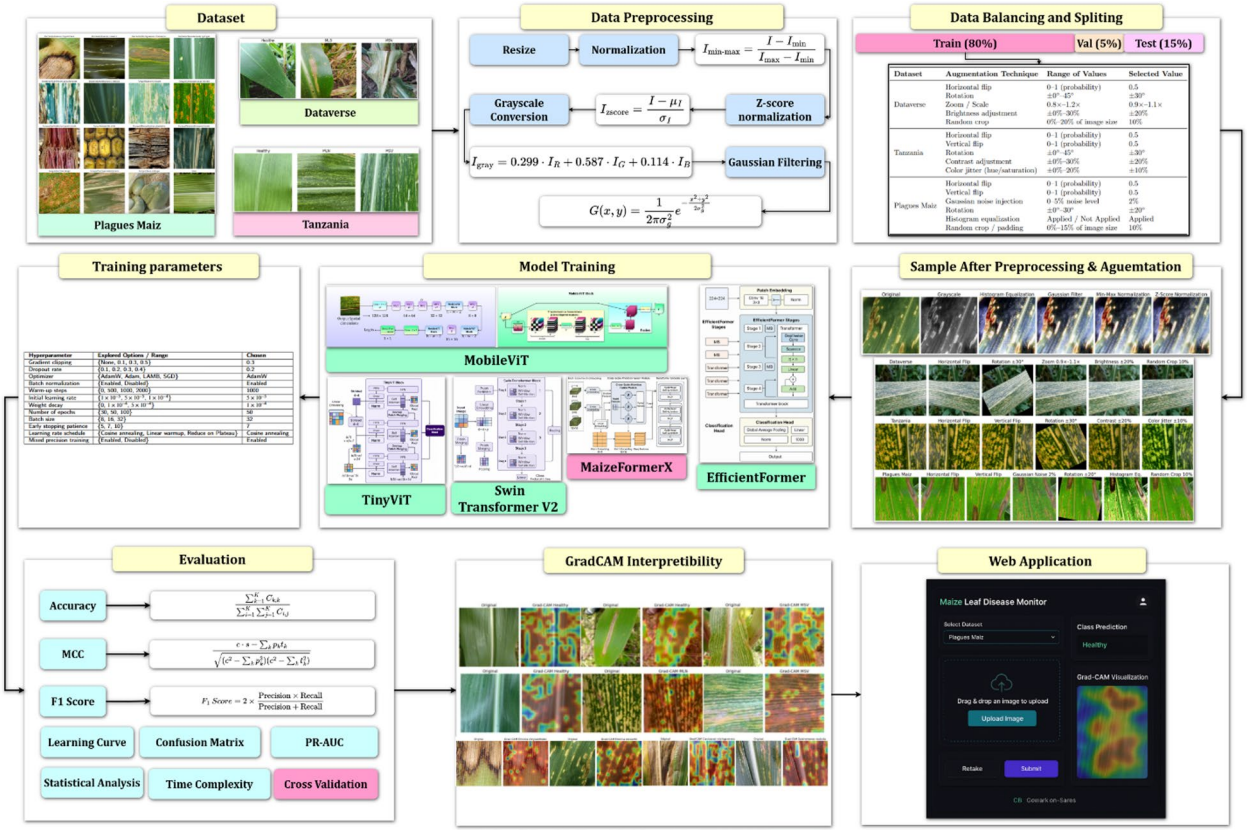


Fig. 1. Overview of the proposed MaizeFormerX classification system.

refinements, MaizeFormerX prioritizes model interpretability and practical usability. It also functions as web-based application that incorporates visual explanations of infected leaf samples.

Key contributions of this study are as follows:

- Proposed a novel lightweight ViT architecture that integrates multi-scale patch embedding and cross-scale attention fusion for accurate and efficient maize leaf disease classification.
- Introduced a dual-scale feature learning strategy that effectively models both fine-grained lesion details and global disease patterns, outperforming evaluated CNN and transformer models.
- Integrating explainable AI (XAI) with Grad-CAM enhances visualizations of model predictions, serving as a valuable qualitative tool for evaluation and future decision-support workflows in agricultural contexts.
- Developed a practical web application for specialists to improve transparency in the rural farming environments.

This paper is structured as follows: Sect. "Related works" reviews related works on maize leaf disease classification using CNN and ViT architectures. Section "Methodology" covers the datasets, preprocessing, and augmentation strategies used in this study, along with the proposed architecture and experimental setup. Section "Result analysis" presents the results, focusing on performance comparisons, cross-domain generalization, and interpretability analyses. Section "Discussion" discusses the key findings, implications, and limitations. Finally, Sectioin "Conclusion" concludes the paper by summarizing the main findings and suggesting future research directions.

Related works

CNN based approaches

CNNs are commonly used for classifying maize leaf diseases due to their ability to extract local features and achieve high accuracy on curated datasets. However, they often struggle to capture global context and generalize effectively under real-world field conditions. Jasrotia et al.[22] proposed a CNN enhanced with Contrast Limited Adaptive Histogram Equalization (CLAHE) and HSV color space conversion, achieving an accuracy of 96.76% on 3,852 PlantVillage images across four classes; however, their method was not validated on diverse datasets. O'Halloran et al.[23] evaluated six CNN architectures on 15,344 images, with EfficientNetV2 B0 achieving an impressive 99.99% accuracy using Grad-CAM for interpretability, but they noted limitations in regional generalizability. Elmasry et al.[24] introduced DenseNetDNN, which combines DenseNet121 with a Deep Neural Network (DNN) classifier, achieving 96.1% accuracy on 4,188 images, although it lacked validation in field conditions. Masood et al.[25] developed MaizeNet, an enhanced Faster R-CNN using ResNet-50 and the

Convolutional Block Attention Module (CBAM), with an accuracy of 97.89% on 2,112 images. However, this model faced longer inference times and issues with false detections in noisy backgrounds.

Malik et al.[26] proposed TreeNet, which combines DenseNet201 and DarkNet-53, achieving 99.8% accuracy on 4,000 images but was also not tested under field conditions. Thakur et al.[27] developed VGG-ICNN, which merges VGG16 and Inception v7 blocks, achieving 91.36% accuracy across five datasets. However, the model's high FLOPs limit its potential deployment, and it lacks field validation. Rajeena et al.[28] used EfficientNet B0 along with Gray Level Co-occurrence Matrix (GLCM) preprocessing, achieving 98.85% accuracy on 3,188 images, but their results were constrained by the dataset size and absence of field validation. Suharto et al.[29] proposed DeCNN with Gaussian denoising, achieving 99% accuracy on 4,000 PlantDoc images; nonetheless, it lacked cross-regional validation and deployment testing.

ViT based approaches

ViTs are increasingly used for plant disease classification due to their ability to capture both global dependencies and fine-grained features. However, they face challenges like high computational requirements and deployment constraints. Ramadan et al.[30] showed that ViT-B/16 achieved 94.51% accuracy on a maize dataset of 4,188 images, but did not validate its performance in real-world conditions. In another study, Ramadan et al.[31] used CycleGAN-based augmentation with DenseNet169 on 4,600 maize leaf images, reaching 98.48% accuracy, but again lacked real-world testing. Isinkaye et al.[32] proposed a VAE-ViT hybrid that achieved 93.2% accuracy on 27,872 images across 12 classes, though it required significant resources. Padshetty et al.[33] developed Twin-ViT with deformable attention and edge-aware modules, achieving 98.88% accuracy on over 17,000 images, which limits its deployment due to computational intensity.

Mayo et al.[34] used ViT and CNN for early disease detection on 26,913 Tanzanian field images, where ViT achieved 93.1% accuracy but lacked cross-regional validation. Gole et al.[35] introduced TrIncNet, a lightweight ViT using Inception modules, with 96.93% accuracy on PlantVillage and field maize images, but no hardware validation was conducted. Li et al.[36] developed Sim-ConvNeXt with SimAM attention and transfer learning, achieving 95.2% accuracy on 17,670 maize leaf images and outperforming Swin-T and ViT, although it requires further real-world optimization. Zhu et al.[37] presented a state-space attention model that achieved 94% accuracy on 7,343 maize leaf images, introducing spatiotemporal attention but facing deployment challenges. Aboelenin et al.[38] created a CNN–ViT hybrid achieving 98% accuracy on PlantVillage images, limited by high computational costs and lacking cross-domain validation. Singh et al.[39] introduced LeafyGAN for GAN-based augmentation, with MobileViT reaching 99.92% on PlantVillage and 75.72% on PlantDoc, improving class balance, but encountering minor artifact issues.

Lightweight models

Researchers have developed lightweight models for detecting maize leaf diseases to address the computational limitations of traditional CNNs and ViTs. Hu et al.[40] proposed an improved YOLOv5s model that incorporates Faster-C3, CoordConv, CARAFE, and channel-wise knowledge distillation. This model achieved a 3.8% improvement over 12,957 annotated images while reducing the number of parameters by 15.5%. However, it experienced slight accuracy trade-offs and limited robustness under diverse field conditions. Li et al.[41] introduced the GhostNet-Triplet-YOLOv8s model, which combines a GhostNet backbone, Triplet Attention, and ECIoU loss. This model achieved 91.4% accuracy on 5,763 images while halving the model size to 11.2 MB. Nonetheless, it faced low recall rates for detecting small lesions. Wang et al.[42] developed ULEN, an ultra-lightweight CNN with 111,000 parameters that uses residual depth-wise convolutions and spatial pyramid pooling. This model achieved a 97.76% F1-score on 3,852 maize images, making it feasible for deployment on Raspberry Pi, although it showed slightly lower accuracy compared to the Xception model. Finally, Ji et al.[43] proposed ICS-ResNet, a lightweight model based on ResNet50 that features improved spatial and channel attention, depth-separable residuals, and cosine annealing. This model achieved 98.87% accuracy on 8,280 maize images while reducing the number of parameters by 69.2%. However, it has not been tested at the field level for real-world validation.

Deep learning has also demonstrated strong effectiveness for plant disease classification across multiple crop domains, reinforcing its value for precision agriculture. For apple leaf disease recognition, Kunduracioglu et al. reported that EfficientNetV2_m achieved 100% accuracy and 100% F1-score, illustrating the capability of efficient CNN backbones to learn highly discriminative symptom features on curated plant datasets[44]. In grape disease/leaf classification, evaluations on a PlantVillage grape subset (4,062 images) and a Grapevine dataset (500 images) showed that several modern deep models can reach very high accuracy, with top configurations reporting 100% accuracy under controlled settings[45]. Similarly, for tomato disease identification, Res2Next50 achieved 99.85% accuracy with an F1-score of 99.82%[46], while for sugarcane leaf disease classification, EfficientNet-b6 achieved 0.9339 accuracy, supporting the practicality of lightweight architectures for agricultural deployment constraints[46]. Collectively, these studies highlight the promise of deep models, but they also underline a recurring limitation: performance is often reported under single-dataset or controlled conditions, leaving cross-domain robustness and interpretable decision support as open challenges, motivating our cross-dataset, lightweight, and explainable MaizeFormerX framework.

Research gap

Recent advancements in maize leaf disease detection using deep learning face several challenges. Many studies use CNNs like DenseNet, MobileNet, and ResNet for local feature extraction, but they struggle with global attention needed for detailed symptom differentiation. ViTs can model globally but are underutilized in maize disease classification, especially in lightweight designs for real-world use. Current models often rely on the PlantVillage dataset, which doesn't reflect real-world lighting and background conditions, leading to poor generalization. Cross-dataset validation is lacking, complicating performance assessment on new data.

Category	Representative prior methods (examples)	Reported strength	Key limitation	How MaizeFormerX addresses the gap
CNN (classification)	CLAHE + HSV CNN [22]	High accuracy on curated images	Limited validation on diverse field settings	Uses heterogeneous datasets + cross-dataset evaluation to quantify domain shift
CNN + XAI	EfficientNetV2 B0 + Grad-CAM[23]	Strong in-dataset accuracy + interpretability	Limited regional generalizability	Reports cross-dataset performance and analyzes explanations across datasets
CNN (limited field testing)	DenseNetDNN[24]	Strong results on limited data	Not evaluated under real farm conditions	Includes real farm-condition Tanzania dataset and domain-shift testing
Detection-style CNN	Faster R-CNN + CBAM (MaizeNet)[25]	Accurate detection in controlled setups	Longer inference + false detections in clutter	Lightweight design with explicit efficiency/latency reporting; CSAF improves context modeling
High-cost CNN hybrids	VGG-ICNN[27]	Multi-dataset evaluation	High FLOPs; deployment constraints	CSAF reduces attention cost via cross-scale interaction; lightweight transformer backbone
Cross-region missing	DeCNN on PlantDoc[29]	High accuracy on PlantDoc	Lacked cross-regional validation	Explicit train-on-source/test-on-target evaluation protocol
Standard ViT	ViT-B/16[30]	Strong global modeling	High compute; limited deployment focus	Efficient multi-scale tokenization + CSAF to inject context at lower attention cost
Heavy ViT variants	Twin-ViT[33]	Competitive accuracy	Computationally intensive	Resource-efficient design validated with Params/FLOPs/latency comparisons
ViT on field images	ViT on Tanzanian field images[34]	Field dataset usage	No cross-regional testing	Adds cross-dataset generalization analysis across three datasets
Lightweight detectors	Improved YOLOv5s[40]	Reduced params + gains	Robustness under diverse field conditions still limited	Uses dataset-specific augmentations + cross-domain evaluation to improve robustness
Ultra-light models	ULEN / ICS-ResNet[42,43]	Very low parameters; edge feasibility	Potential accuracy drop / limited validation	Balances efficiency + accuracy and validates across multiple acquisition domains

Table 1. Structured comparison of prior maize leaf disease methods, key limitations, and how MaizeFormerX addresses them.



Fig. 2. Sample image from each class of the Dataverse dataset.

While some hybrid CNN-ViT approaches and attention mechanisms exist, they tend to increase complexity and computational demands, making them impractical for low-resource devices. Explainability also remains an issue, with limited use of Grad-CAM. Table 1 summarizes representative state-of-the-art approaches, their recurring limitations, and how MaizeFormerX directly addresses these gaps.

To address these issues, we introduce MaizeFormerX, a hybrid transformer-based architecture that combines ViTs' global modeling with multi-scale patch embeddings and Cross-Scale Attention Fusion (CSAF) for effective recognition under real-world constraints. We validate it through cross-dataset evaluations on three diverse maize leaf datasets, demonstrating strong generalization under domain shift. Additionally, we integrate Grad-CAM into a web application for interactive visualization of model predictions, enabling transparent and actionable maize disease identification.

Methodology

Recent systematic evidence confirms that deep learning has become the dominant approach for plant disease classification, consistently outperforming traditional handcrafted pipelines, while also highlighting practical gaps such as dataset bias, limited domain generalization, and deployment constraints in field conditions[47]. Building on these observations, our methodology emphasizes robust cross-dataset evaluation, deployment-oriented efficiency, and explainability to support reliable maize disease classification in real-world settings.

Data description

We utilized three publicly available datasets for classifying maize leaf diseases: the Dataverse, Tanzania, and Plagues Maiz datasets. These datasets were chosen to encompass a broad range of disease manifestations, environmental conditions, and pathogen types. Importantly, the datasets were used independently and were not merged across sources, since they follow different class taxonomies (e.g., MLB vs. MLN vs. pathogen-level subclasses) and exhibit distinct acquisition conditions. The Dataverse dataset[48](shown in Fig. 2) includes a total of 15,822 images divided into three categories: Healthy (5,327 images), MLB (5,279 images), and MSV (5,216

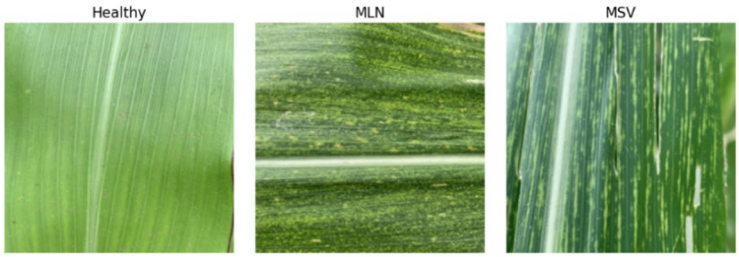


Fig. 3. Sample image from each class of the Tanzania dataset.

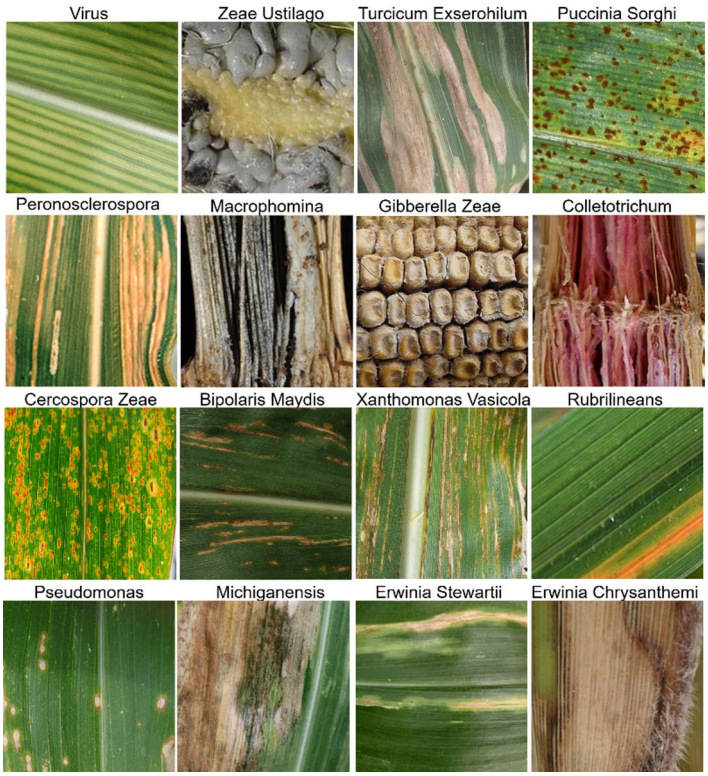


Fig. 4. Sample image from each class of the Plagues Maiz dataset.

images). This dataset features a balanced distribution of healthy and diseased leaf images, making it suitable for training models with reduced class bias. The Tanzania dataset[49](see Fig. 3) consists of 9,356 images collected from real farm conditions across Tanzania and contains three categories: Healthy (3,073 images), MLN (3,231 images), and MSV (3,052 images). This dataset introduces region-specific variability in lighting, background clutter, and symptom appearance, enhancing the evaluation of robustness across diverse agroecological zones.

The Plagues Maiz dataset[50] consists of 850 images and provides detailed annotations across 16 microbial subclasses (see Fig. 4). It includes three major pathogen groups: Bacteria with six subclasses (Erwinia chrysanthemi, Erwinia stewartii, Clavibacter michiganensis, Pseudomonas syringae, Rubrilineans pseudomonas, and Xanthomonas vasicola), Fungi with seven subclasses (Bipolaris maydis, Cercospora zeae-maydis, Colletotrichum graminicola, Gibberella zeae, Macrophomina phaseolina, Peronosclerospora sorghi, Puccinia sorghi, Exserohilum turcicum, and Ustilago zeae), and Viruses with 50 samples. Despite its relatively smaller scale, Plagues Maiz offers rich pathogen-level granularity, making it valuable for assessing fine-grained recognition capabilities.

Data preprocessing and augmentation

All images were resized to 224×224 pixels to match the input requirements of the pretrained ViT backbone. Images were converted to RGB format when necessary. Two normalization strategies were considered to standardize the input data. First, min-max normalization rescales pixel intensities to the [0,1] range, as described in Eq. 1, where I is the input pixel intensity, and I_{min} and I_{max} are the minimum and maximum pixel intensities in the image.

$$I_{\text{min-max}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \tag{1}$$

Second, z-score normalization standardizes pixel values relative to the dataset distribution, as shown in Eq. 2, where μ_I and σ_I represent the mean and standard deviation of the pixel intensities, respectively.

$$I_{\text{zscore}} = \frac{I - \mu_I}{\sigma_I} \tag{2}$$

Additional preprocessing methods were included to improve image quality and feature extraction. Grayscale conversion, given in Eq. 3, was applied to highlight texture features, where I_R, I_G, and I_B are the red, green, and blue channel intensities.

$$I_{\text{gray}} = 0.299 \cdot I_R + 0.587 \cdot I_G + 0.114 \cdot I_B \tag{3}$$

Histogram equalization was used to enhance contrast under variable lighting conditions in Eq. 4, where L is the number of intensity levels, p_j is the probability of intensity level j, and $I_{\text{eq}}(k)$ is the new mapped intensity value.

$$I_{\text{eq}}(k) = \text{round}\left((L-1)\sum\nolimits_{j=0}^{k} p_j\right) \tag{4}$$

Gaussian filtering was employed to reduce noise while preserving edge information shown in Eq. 5, where σ_g is the Gaussian standard deviation controlling the amount of smoothing. The sample image after preprocessing is shown in Fig. 5.

$$G(x,y) = \frac{1}{2\pi\sigma_g^2} e^{-\frac{x^2+y^2}{2\sigma_g^2}} \tag{5}$$

Data augmentation was applied only to the training sets to improve generalization while keeping validation and test sets intact. We used geometric (flips, rotations, zooms), photometric (brightness, contrast, color jitter), and noise-based (Gaussian injection) transformations, as detailed in Table 2. This approach simulated real-world variability and minimized overfitting.

To tackle class imbalance in the Plagues Maiz dataset, which had an initial imbalance ratio of about 17:1, we employed stratified, class-conditional augmentation with expansion factors of $\times 2$, $\times 4$, $\times 6$ for underrepresented classes, resulting in a post-augmentation ratio of roughly 1:1. The effective sample size for the minority class c was calculated as Eq. 6. This approach avoided synthetic oversampling while preserving semantic validity and diversity. The augmentation pipeline was integrated on-the-fly within the training loop to reduce memory overhead and enable dynamic mini-batches. Additionally, the individual effects of each augmentation technique are illustrated in Fig. 6.

$$N_{eff}^{(c)} = N_{orig}^{(c)} \times \alpha_c, \quad \alpha_c \in \{2, 4, 6\} \tag{6}$$

Photometric normalization and dataset-agnostic augmentations were further employed to reduce dataset-specific biases (e.g., between Dataverse and Tanzania), enhancing domain generalization. While fairness metrics were not calculated, the use of diverse datasets and neutral augmentations aimed to enhance data variability. However, the potential impact on reducing bias in real-world applications still needs to be validated through dedicated fairness assessments and subgroup analyses.

For all datasets, we performed independently splitting into training (80%), validation (5%), and testing (15%) sets. Augmentation was applied only to the training split to prevent information leakage. In addition to in-dataset testing, we report cross-dataset generalization by training on one dataset and directly testing on the others without re-training/fine-tuning (Table 10). Since label sets are not identical across datasets, cross-dataset evaluation follows a shared-label protocol, restricting evaluation to the intersection of classes common to the source and target datasets (e.g., Healthy and MSV shared by Dataverse and Tanzania). Detailed class-wise

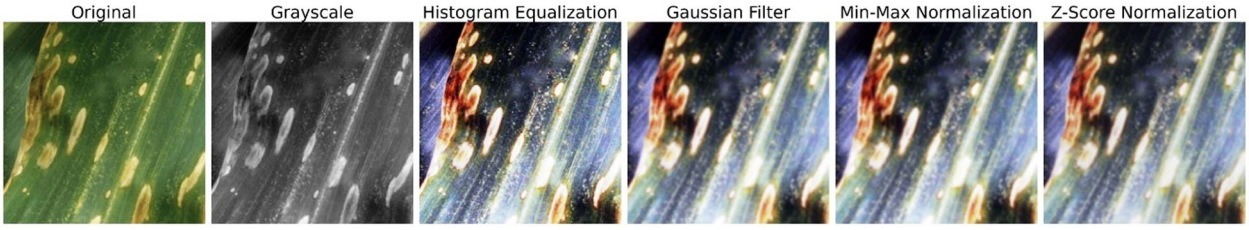


Fig. 5. Visualization of preprocessing techniques applied to the maize leaf image. From left to right: (1) original image, (2) min-max normalization, (3) z-score normalization, (4) grayscale conversion, (5) histogram equalization applied to RGB channels, and (6) Gaussian filtering.

Dataset	Augmentation Technique	Range of Values	Selected Value
Dataverse	Horizontal flip	0–1 (probability)	0.5
	Rotation	± 0°–45°	± 30°
	Zoom / Scale	0.8×–1.2×	0.9×–1.1×
	Brightness adjustment	± 0%–30%	± 20%
	Random crop	0%–20% of image size	10%
Tanzania	Horizontal flip	0–1 (probability)	0.5
	Vertical flip	0–1 (probability)	0.5
	Rotation	± 0°–45°	± 30°
	Contrast adjustment	± 0%–30%	± 20%
	Color jitter (hue/saturation)	± 0%–20%	± 10%
Plagues Maiz	Horizontal flip	0–1 (probability)	0.5
	Vertical flip	0–1 (probability)	0.5
	Gaussian noise injection	0–5% noise level	2%
	Rotation	± 0°–30°	± 20°
	Histogram equalization	Applied / Not Applied	Applied
	Random crop / padding	0%–15% of image size	10%

Table 2. Summary of augmentation techniques, parameter ranges, and selected values for each dataset.

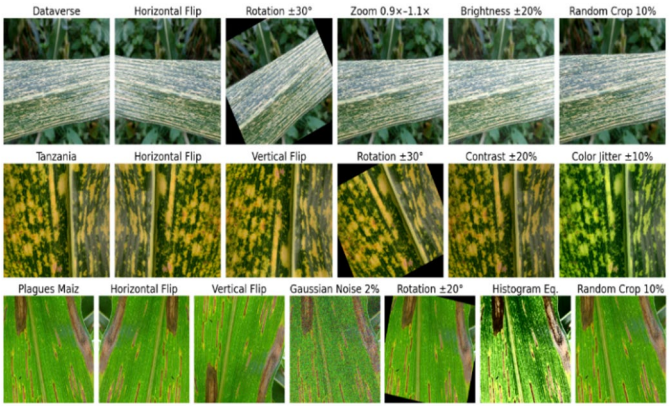


Fig. 6. Original image and individual augmentation outputs applied to maize leaf samples.

distributions and augmentation scaling are provided in Table 3, ensuring transparent, reproducible preparation for maize leaf disease classification.

Experimental models

MobileViT

It is a lightweight hybrid ViT architecture optimized for efficient image classification[51]. It integrates CNNs for local feature extraction and transformer-based modules for capturing long-range dependencies, making it well-suited for fine-grained tasks such as maize leaf disease classification. As shown in Fig. 7, the architecture begins with a 3×3 convolutional layer to extract low-level features from the input image, followed by a series of MobileNetV2 (MV2) blocks that progressively downsample the spatial resolution ($128 \times 128 \rightarrow 64 \times 64 \rightarrow 32 \times 32 \rightarrow 16 \times 16 \rightarrow 8 \times 8$). Between these MV2 blocks, MobileViT blocks are inserted to enhance the representation capacity by combining local and global context. Each MobileViT block includes three key components: local feature extraction via 3×3 convolutions, global feature modeling through transformer layers applied to non-overlapping patches, and feature fusion using 1×1 convolutions and a spatial fusion layer:

$$\mathrm{f_{local}} = \mathrm{Conv3 \times 3\,(X)}\,, \quad \mathrm{f_{global}} = \mathrm{Transformer\,(Unfold\,(f_{local}))}\,, \quad \mathrm{Y = Fusion\,(f_{local}, f_{global})}$$

Unlike traditional ViTs, it unfolds convolutional features dynamically, preserving spatial locality essential for lesion boundary identification. With fewer than 5 M parameters, MobileViT supports multi-scale hierarchies, enabling accurate maize disease classification with low computational cost. Its compact design makes it suitable for agricultural applications.

Dataset	Class	Original Count	Training (80%)	Target 2×	Target 4×	Target 6×	Validation (5%)	Testing (15%)
Dataverse	Healthy	5327	4261	8522	17,044	25,566	266	800
	MLB	5279	4223	8446	16,892	25,338	264	792
	MSV	5216	4172	8344	16,688	25,032	261	783
	Total	15,822	12,656	25,312	50,624	75,936	791	2,375
Tanzania	Healthy	3073	2460	4920	9840	14,760	154	459
	MLN	3231	2585	5170	10,340	15,510	162	484
	MSV	3052	2441	4882	9764	14,646	153	458
	Total	9356	7486	14,972	29,944	44,916	469	1,401
Plagues Maiz	Erwinia chrysanthemi	50	40	80	160	240	3	7
	Erwinia stewartii	50	40	80	160	240	3	7
	Clavibacter michiganensis	51	41	82	164	246	3	7
	Pseudomonas syringae	51	41	82	164	246	3	7
	Rubrilineans pseudomonas	51	41	82	164	246	3	7
	Xanthomonas vasicola	50	40	80	160	240	3	7
	Bipolaris maydis	50	40	80	160	240	3	7
	Cercospora zeae-maydis	50	40	80	160	240	3	7
	Colletotrichum graminicola	50	40	80	160	240	3	7
	Gibberella zeae	51	41	82	164	246	3	7
	Macrophomina phaseolina	50	40	80	160	240	3	7
	Peronosclerospora sorghi	50	40	80	160	240	3	7
	Puccinia sorghi	51	41	82	164	246	3	7
	Exserohilum turcicum	50	40	80	160	240	3	7
	Ustilago zeae	51	41	82	164	246	3	7
	Virus	50	40	80	160	240	3	7
	Total	850	680	1,360	2,720	4,080	51	119

Table 3. Class distribution and data splits across all datasets.

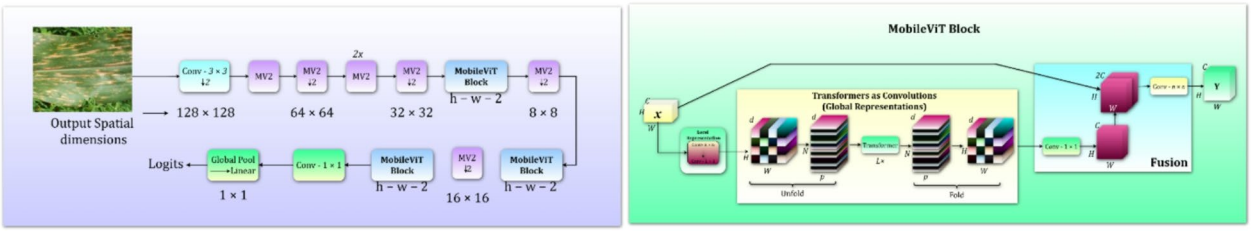


Fig. 7. The left panel shows the overall pipeline combining convolutions, MV2 units, and MobileViT blocks for feature extraction and classification. The right panel details a MobileViT block, fusing local features with global representations learned via transformer-based processing.

EfficientFormer

It is a lightweight hybrid ViT that balances accuracy and efficiency for tasks like maize leaf disease classification[52]. The architecture integrates MobileNet-inspired bottleneck (MB) blocks for local feature extraction and transformer modules for global attention. Input images (224 × 224) are embedded via a 3 × 3 convolution and normalization layer. The first three stages use MB blocks with depthwise separable convolutions and squeeze-excitation units to extract fine-grained features (see Fig. 8). The final stage uses transformer blocks with spatial refinement, channel reduction, and linear transformation $\eta \cdot \mathrm{n}$. Residual additive fusion is used to enhance gradient flow and stabilize training:

$$Z = \mathrm{Norm}\left(\mathrm{Conv3x3}\left(X\right)\right), \quad Y = X + \mathrm{Linear}\left(\eta{\cdot}\mathrm{Squeeze}\left(\mathrm{DepthwiseConv}\left(X\right)\right)\right)$$

The classification head uses global average pooling, normalization, and a linear layer to generate class logits. By applying self-attention only in the final stage, The model processes semantically rich features with minimal overhead. Its modular hybrid design supports efficient local-to-global representation learning, providing a strong balance between accuracy and latency.

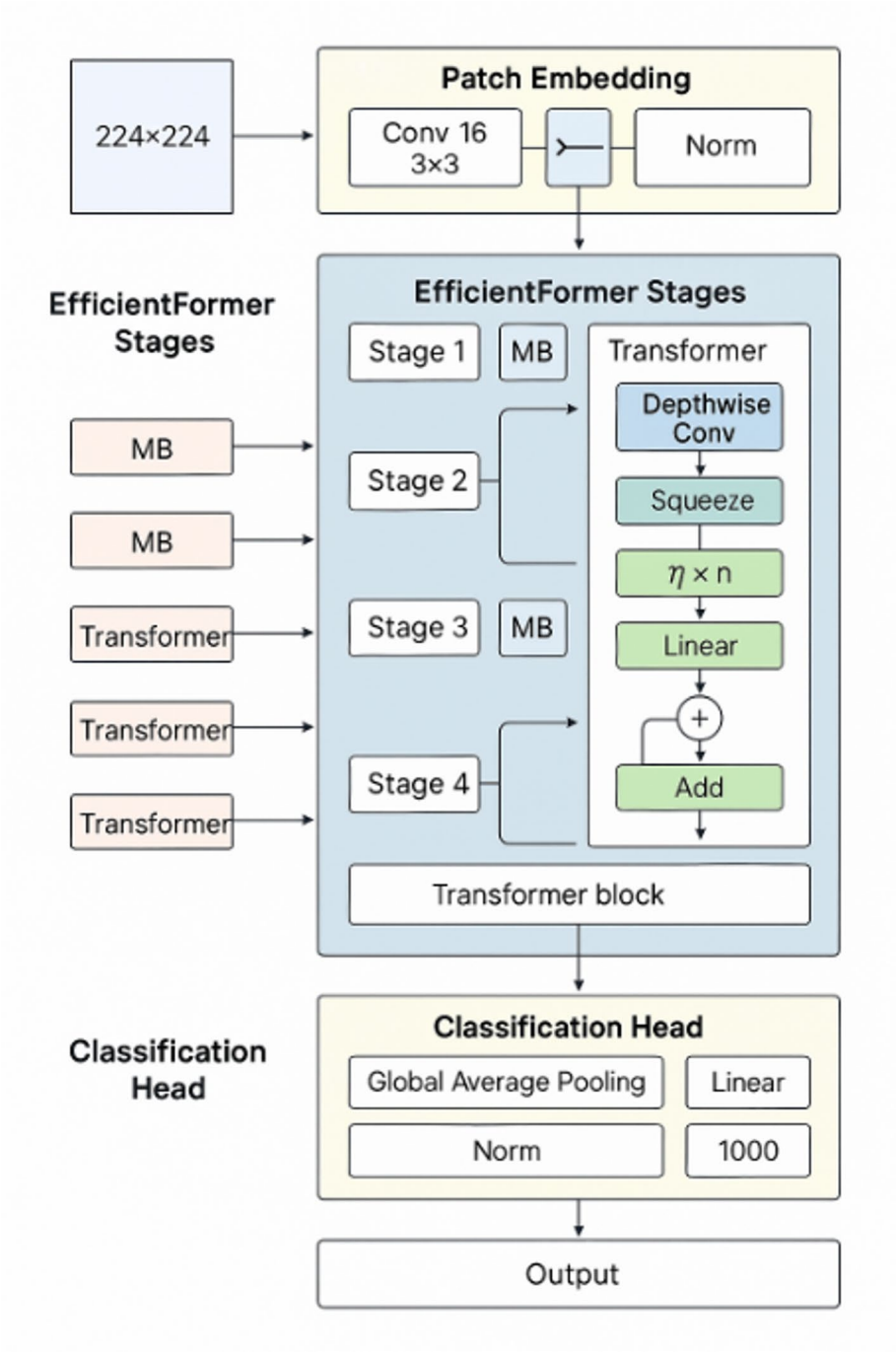


Fig. 8. Overview of the EfficientFormer architecture. It includes patch embedding, four-stage blocks with MBConv and lightweight transformers, followed by a classification head using global average pooling and a linear layer for final prediction.

TinyViT

It is an efficient ViT that combines CNNs' spatial biases with transformers' global modeling abilities, which is vital for capturing fine-grained lesion textures and long-range dependencies. As shown in Fig. 9, the model begins with a linear embedding layer, followed by two strided convolutional layers using a 4×4 kernel to lower spatial resolution and create hierarchical feature maps. These maps are processed through TinyViT blocks at various resolutions for multi-scale representation learning[53], capturing both localized patterns and broader disease spread. Each block includes a linear projection, normalization, low-rank self-attention for global context, and a feed-forward network for enhancement. Instead of traditional tokenization, it uses an overlapping patch merging module to retain crucial high-frequency spatial cues[54]. The process can be formally represented as:

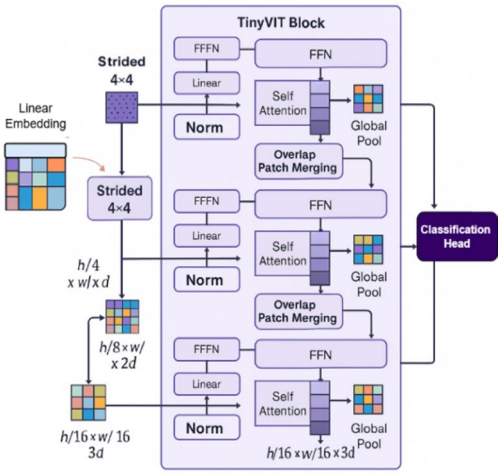


Fig. 9. Architecture of the TinyViT model. It starts with strided linear embedding, followed by stacked TinyViT blocks combining feedforward networks, self-attention, normalization, and overlap patch merging. Multi-scale features are extracted and globally pooled before being passed to the classification head for final prediction.

$$E = \text{Linear}\left(\text{Patch}\left(X\right)\right), \quad A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V, \quad M = \text{ConvOverlap}\left(A\right)$$

This architecture ensures efficient global modeling with reduced computational overhead, and the initial convolutional embeddings provide strong spatial priors, boosting generalization in real-world agricultural imaging scenarios affected by environmental and domain shifts.

Swin transformer V2

This architecture is well-suited for maize leaf disease classification due to its hierarchical design, windowed multi-head self-attention (W-MSA), and spatially adaptive token aggregation[55]. It enables fine-grained localization of pathological cues (e.g., chlorotic speckling, interveinal discoloration, necrotic lesion clustering) while preserving global structural coherence necessary for understanding disease progression. As shown in Fig. 10, given an input image $\mathrm{I} \in \mathrm{R}^{\mathrm{H}_0 \times \mathrm{W}_0 \times 3}$ the model applies non-overlapping patch tokenization followed by a linear projection into $\mathrm{E}_0 \in \mathrm{R}^{\frac{\mathrm{H}_0}{\mathrm{P}_0} \times \frac{\mathrm{W}_0}{\mathrm{P}_0} \times \mathrm{D}_0}$. It progresses through four stages, each employing patch merging to hierarchically fuse adjacent tokens:

$$\mathrm{E}_{\mathrm{u,v}}^{\mathrm{s}+1} = \mathrm{W}_{\mathrm{pm}} \cdot \left[\mathrm{E}_{\mathrm{2u,2v}}^{\mathrm{s}}, \mathrm{E}_{\mathrm{2u+1,2v}}^{\mathrm{s}}, \mathrm{E}_{\mathrm{2u,2v+1}}^{\mathrm{s}}, \mathrm{E}_{\mathrm{2u+1,2v+1}}^{\mathrm{s}}\right]$$

where W_{pm} is a learned linear transformation matrix, and (u, v) denotes spatial indices in stage s. This token downsampling mechanism facilitates efficient multiscale feature encoding and progressively increases the effective receptive field. Each transformer block within a stage incorporates pre-norm residual learning via LayerNorm:

$$\text{LayerNorm}\left(F\right) = \frac{F - \mu_F}{\sqrt{\sigma_F^2 + \epsilon}} \cdot \alpha + \beta$$

Here μ_{F}, σ_{F}^2, α, and β represent the channel-wise statistics and learnable affine parameters, ensuring gradient flow stability and normalization-aware optimization. The key component, Shifted W-MSA, alternates attention windows across layers to efficiently capture cross-window dependencies. Each attention head is augmented with a learnable relative position bias:

$$\mathrm{R}_{\mathrm{p,q}} = \psi\left(\mathrm{p} - \mathrm{q}\right)$$

Here, (ψ) is a parameterized function mapping relative positional differences between patches (p) and (q), enriching spatial inductive bias to aid precise lesion boundary identification.

The model serves as a high-capacity, inductive-biased vision backbone capable of modeling localized pathology under varied imaging conditions. Its core advantage lies in unifying linear computational scalability with hierarchical multiscale tokenization, enabling long-range dependency modeling via sparse attention while preserving spatial resolution through non-uniform patch-wise feature fusion. The integration of dynamic window shifting, relative positional embeddings, and reparameterized normalization layers renders it highly

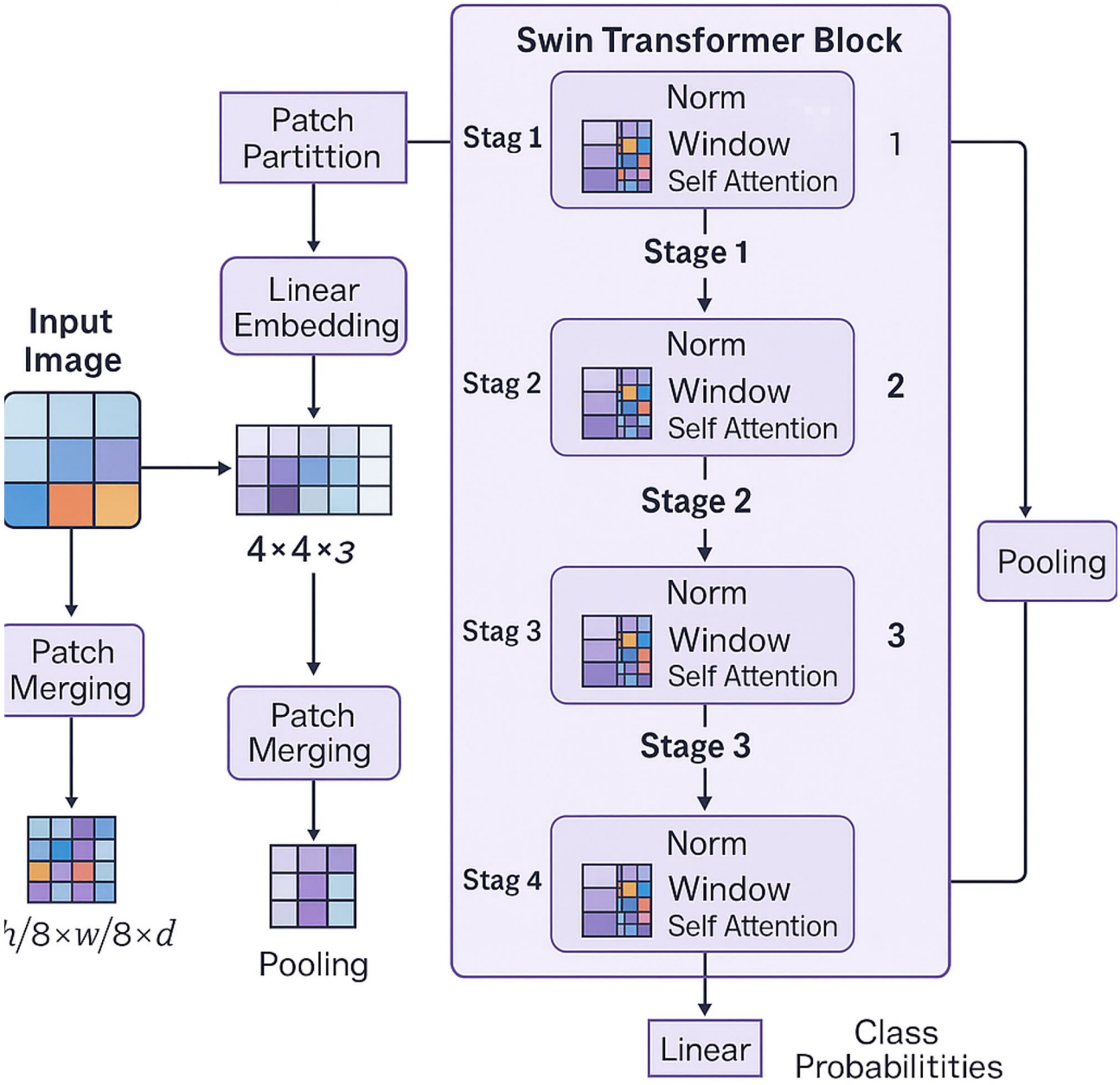


Fig. 10. Swin Transformer V2 architecture with hierarchical stages. It includes patch partitioning, linear embedding, window-based self-attention, patch merging for downsampling, and a final linear layer.

effective in capturing both low-level stochastic lesion signatures and high-level semantic regions, such as necrosis zones, fungal spore patterns, and edge-based chlorosis in maize foliage.

Proposed MaizeFormerX

Problem formulation. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ denote a maize leaf image dataset, where $x_i \in \mathbb{R}^{H\times W\times 3}$ is an RGB image and $y_i \in \{1, \ldots, C\}$. is the disease label. MaizeFormerX defines a classifier f_Θ that maps an input image to logits $z = f_{\boldsymbol{\Theta}}(x) \in \mathbb{R}^C$. and class probabilities $p_\Theta(y|x) = \mathrm{softmax}(z)$. The model is trained by minimizing cross-entropy with weight decay:

$$\min_{\Theta} \frac{1}{N} \sum_{i=1}^{N} \mathrm{CE}\left(p_\Theta(\cdot \mid x_i), y_i\right) + \lambda \parallel \Theta \parallel_2^2.$$

introduce MaizeFormerX, a hybrid transformearchitecture designed for effective maize leaf disease classification. The complete training–inference pipeline is summarized in Algorithm 1, including dataset splitting, dataset-

specific augmentation, model construction, optimization, and evaluation. As illustrated in Fig. 11, MaizeFormerX integrates multi-scale patch embedding, Cross-Scale Attention Fusion (CSAF), and a transformer encoder stack, enabling the model to jointly capture fine lesion textures .nd global contextual cues required for accurate disease identification.

Multi-scale patch embedding. Given an in.ut image $x \in \mathbb{R}^{H\times W\times 3}$, MaizeFormerX extracts non-overlapping patches at three resolutions $p \in \{4, 8, 16\}$. Each patch set is linearly projected into a unified embedding space to preserve both localized patterns (e.g., necrotic spots, rust pustules) and broader structural spread (e.g., chlorosis, lesion coalescence). The resulting token embeddings are.

$X^{(4)} \in \mathbb{R}^{N_4\times D}, X^{(8)} \in \mathbb{R}^{N_8\times D}, X^{(16)} \in \mathbb{R}^{N_{16}\times D}$,

where $N_s = HW/s^2$ and D denotes the token dimension.

Cross-scale attention fusion (CSAF). The extracted tokens then flow into CSAF, which is the core innovation of MaizeFormerX. Unlike conventional multi-scale fusion mechanisms (e.g., FPN or U-Net) that combine features using fixed addition/concatenation after resizing, CSAF performs content-adaptive token alignment across resolutions using multi-head cross-attention. Concretely, fine-scale tokens query coarser-scale tokens that act as a compact semantic memory. Using $X^{(4)}$as queries and $X^{(16)}$ as keys/values, CSAF computes:

$$Q = X^{(4)}W_Q, K = X^{(16)}W_K, V = X^{(16)}W_V,$$

$$\text{Attn}\left(X^{(4)}?X^{(16)}\right) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

where $W_Q, W_K, W_V \in \mathbb{R}^{D\times d}$ are learnable projections and d is the per-head dimension. The fused representation is formed via residual fusion with normalization:

$$F_{\text{fused}} = \text{LN}\left(X^{(4)} + \text{Attn}\left(X^{(4)}?X^{(16)}\right)\right),$$

(followed by an additional lightweight fusion stage using $X^{(8)}$ as an intermediate memory). This design injects global context into fine lesion tokens while preserving spatial precision, producing context-aware multi-resolution embeddings. CSAF is computationally efficient because it replaces expensive dense attention on the finest grid with cross-scale attention against a much smaller token set. Full self-attention at the finest scale scales as $\mathcal{O}\left(N_4^2 D\right)$, whereas CSAF scales as $\mathcal{O}\left(N_4 N_{16} D\right)$, where $N_{16} \ll N_4$. In contrast to CoTNet, which injects contextual attention primarily within convolutional neighborhoods, CSAF explicitly models cross-resolution correspondence in token space, enabling global-context refinement with reduced computational overhead.

Transformer encoder and classifier. The fused tokens are processed by a transformer encoder stack of L layers (Algorithm 1), where each layer includes Multi-Head Self-Attention (MHSA) and a feed-forward MLP. The encoder captures long-range dependencies across the fused token space and enhances discriminative power. The final encoded representation F_L is aggregated using Global Average Pooling and mapped to class probabilities:

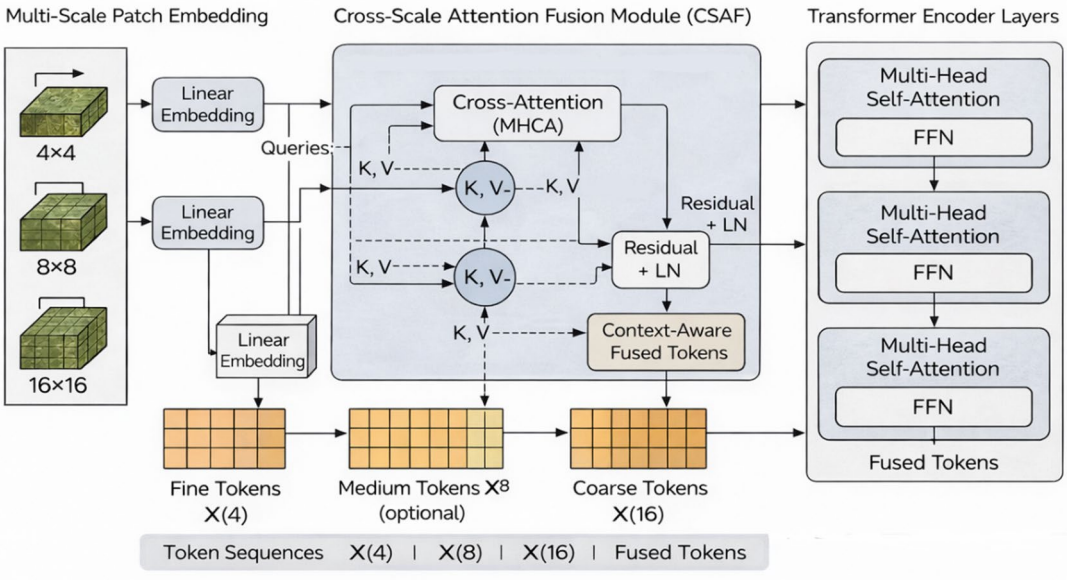


Fig. 11. Architecture of MaizeFormerX. Multi-scale token sets $X^{(4)}, X^{(8)}, X^{(16)}$ are extracted via patch embeddings and linearly projected. CSAF computes cross-scale fusion using multi-head cross-attention $\text{Attn}(X^{(4)} \leftarrow X^{(16)})$, where $X^{(4)}$ provides queries and $X^{(16)}$ provides keys/values, yielding context-aware fine tokens through residual fusion and normalization. The fused tokens are forwarded to stacked transformer encoder layers for representation learning and disease classification.

Require: Image dataset $D = (x_i, y_i)_{i=1}^{N}$ with C classes
Ensure: Trained model M; predictions ŷ; evaluation metrics
— Data Preparation —
Split D into D_{train}: 80%, D_{val}: 5%, D_{test}: 15%
for all x ∈ D_{train} do
  Resize x to 224×224, convert to RGB
  Apply dataset-specific augmentations with factor α ∈ {2, 4, 6} (on-the-fly)
end for
— Model Construction —
function MAIZEFORMERX(x) ▷ Multi-scale patch embedding
  for all patch size p ∈ {4, 8, 16} do
    Extract non-overlapping patches P_p
    $E_p \leftarrow Linear(P_p)$
  end for
  $F_0 \leftarrow CSA.Fuse(E_p)$ ▷ Cross-scale attention fusion (concatenation + cross-attention)
  for ℓ = 1 to L do ▷ Transformer encoder stack of L layers
    $F_\ell \leftarrow MHSA(F_{\ell-1}) + MLP(F_{\ell-1})$
  end for
  z ← GlobalAvgPool(F_L)
  ŷ ← Softmax($Wz + b$)
  return ŷ
end function
— Training Loop —
for epoch = 1 to E_{max} do
  for all (x, y) ∈ D_{train} do
    ŷ ← MAIZEFORMERX(x) ▷ Forward pass
    $L \leftarrow CE(y, \hat{y}) + \lambda||\Theta||_2^2$ ▷ Cross-entropy + weight decay (λ)
    $\Theta \leftarrow \Theta - \eta\nabla_\Theta L$ ▷ Adam with cosine annealing (η)
  end for
  if validation loss has not improved for 4 epochs then
    break
  end if ▷ Early stopping
end for
— Evaluation —
Compute classification metrics (Accuracy, F1, PR AUC, MCC) on D_{test}
return trained model M, predictions ŷ, evaluation metrics

Algorithm 1. Proposed MaizeFormerX for maize leaf disease classification

$$z \leftarrow \text{GlobalAvgPool}\left(F_L\right), \widehat{y} = \text{Softmax}(Wz + b).$$

During training, the model is optimized using cross-entropy with weight decay regularization,

$$\mathcal{L} = \text{CE}(\widehat{y}, y) + \lambda \parallel \Theta\parallel_2^2,$$

and updated using Adam with cosine annealing as specified in Algorithm 1.

MaizeFormerX outperforms traditional single-scale CNN and transformer baselines by effectively encoding spatial hierarchies and semantic dependencies across scales. The multi-resolution embeddings preserve crucial texture details and structural variations, while CSAF adaptively merges low-level lesion cues with high-level context through cross-attention, producing spatially accurate and semantically meaningful representations. The encoder stack, with layer normalization, residual connections, and GELU-activated FFN layers, further strengthens inter-class separability and improves generalization across acquisition domains.

Hyperparameter optimization for model training

Experiments were carried out on a workstation with an NVIDIA RTX A6000 GPU (48 GB VRAM), Intel Core i9 processor, and 128 GB RAM. This setup allowed for efficient batch processing, multi-scale patch embeddings, and cross-domain validation. Model development and experiments utilized Python 3.10, PyTorch, and CUDA 12.1 for scalable computation. We implemented a carefully designed hyperparameter optimization strategy, summarized in Table 3. A learning rate of 5×10^{-5} provided the best balance between convergence stability and training speed, avoiding the instability of larger values and the slowness of smaller ones. A batch size of 32 offered the optimal trade-off between gradient stability and GPU utilization. For regularization, a dropout rate of 0.3 effectively reduced overfitting, while the Adam optimizer consistently provided robust convergence. A weight decay of 5×10^{-5} maintained generalization without excessively suppressing the learning signal. Among learning rate schedules, cosine annealing proved most effective, and 300 warm-up steps helped stabilize early training. We used a maximum of 30 epochs to prevent underfitting while avoiding diminishing returns. Early stopping with a patience of 4 further ensured stable and efficient training. Table 4 summarizes the hyperparameter settings.

Hyperparameter	Range/Candidates	Selected value
Learning rate	{1e-6, 1e-5, 5e-5, 1e-4}	5×10^{-5}
Batch size	{16, 32, 64}	32
Dropout rate	{0.1, 0.3, 0.4}	0.3
Optimizer	{SGD, Adam, RMSprop}	Adam
Weight decay	{1e-6, 5e-5, 1e-4}	5×10^{-5}
Learning rate schedule	{Constant, Linear Decay, Cosine Annealing}	Cosine Annealing
Warm-up steps	{0, 100, 300}	300
Max Epochs	{20, 30, 40}	30
Early stopping patience	{3, 4, 6}	4

Table 4. Hyperparameter search ranges and selected values.

Evaluation metrics

To assess the performance of our model in multiclass maize leaf disease classification, we utilized four evaluation metrics: Micro Accuracy, Micro F1-score, Micro Precision-Recall AUC (PR AUC), and the Matthews Correlation Coefficient (MCC). Micro Accuracy measures the proportion of correctly predicted samples across all classes, treating each instance equally. It is calculated as follows Eq. 1, where $C_{i,j}$ denotes the number of samples from actual class i that have been predicted as class j, and $C_{k,k}$ indicates the number of correct predictions. Micro F1-score represents the harmonic mean of micro-averaged precision and recall, providing a balanced evaluation that is particularly useful in the presence of class imbalance. It is defined as Eq. 2, where TP_i, FP_i, and FN_i refer to the true positives, false positives, and false negatives for class i, respectively.

$$\text{Micro Accuracy} = \frac{\sum_{k=1}^{K} C_{k,k}}{\sum_{i=1}^{K} \sum_{j=1}^{K} C_{i,j}} \tag{1}$$

$$\text{MicroF1} = \frac{\sum_{i=1}^{n} (2 \cdot TP_i)}{\sum_{i=1}^{n} (2 \cdot TP_i + FP_i + FN_i)} \tag{2}$$

Micro PR AUC evaluates the trade-off between precision and recall across various thresholds, offering insights into the model's discriminatory capabilities within a multiclass context. To measure predictive balance, we calculated the MCC for both binary and multiclass tasks. The formulas for MCC as Eqs. 3 and 4, where c represents the total number of correct predictions, s indicates the total number of samples, p_k is the predicted count, and t_k is the true count for class k.

$$\text{MCC} = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)\,(TP + FN)\,(TN + FP)\,(TN + FN)}} \tag{3}$$

$$\text{MCC}_{\text{multi-class}} = \frac{c \cdots - \sum_k p_k t_k}{\sqrt{\left(c^2 - \sum_k p_k^2\right)\left(c^2 - \sum_k t_k^2\right)}} \tag{4}$$

We split each dataset into 80% training, 5% validation, and 15% testing, preserving class proportions. Augmentation was applied only to the training split, while validation and test sets remained untouched. To ensure evaluation stability, we repeated the full training–validation–testing procedure using multiple random seeds (e.g., {42, 77, 123, 2025, 9090}) and report results as mean ± standard deviation across runs. Accordingly, the shaded region in the learning curves represents the ± 1 standard deviation band across seeds.

Result analysis

Dataset-wise performance comparison of experimental models

Table 5 clearly shows that MaizeFormerX outperforms other models across all datasets and augmentation levels. On Dataverse, it leads with 96.4 ± 0.9% accuracy at ×2, surpassing Swin Transformer V2, the strongest baseline, by + 2.4 points. The gap widens with higher augmentation, most notably at ×6, where MaizeFormerX achieves 97.8 ± 0.8%, while Swin V2 and TinyViT remain lower at 95.0 ± 0.9% and 94.8 ± 1.0%, respectively. This trend is echoed in discrimination metrics, with MaizeFormerX showing superior class separability. On the more challenging Tanzania split, MaizeFormerX consistently ranks highest at all augmentation levels, starting at 96.9 ± 0.9% accuracy at ×2, compared to Swin V2's 94.2 ± 1.0%. It peaks at 97.5 ± 0.9% at ×4, maintaining a ~ 3.3-point lead. Even at ×6, where others struggle, it remains strong at 95.5 ± 1.1%, far outperforming MobileViT (86.8 ± 1.6%) and EfficientFormer (89.3 ± 1.5%). On Plagues Maiz, MaizeFormerX again leads with scores of 95.4 ± 1.0% (×2), 96.9 ± 0.9% (×4), and 95.7 ± 0.9% (×6). In contrast, Swin V2 peaks at 94.5 ± 1.0% (×4) and TinyViT stays around 92.8 ± 1.2%. This suggests MaizeFormerX excels in extracting disease-discriminative features, not just benefiting from augmentation.

Table 6 presents mobile CNN baselines. Although augmentation helps some CNNs, their performance varies by dataset and overall remains lower. The best CNN, EfficientNet-B1, achieves 93.4 ± 1.0% accuracy at ×6 on

Augmentation	Metric	MobileViT	EfficientFormer	TinyViT	Swin Transformer V2	MaizeFormerX
Dataverse						
×2	Accuracy	89.3 ± 1.4	90.5 ± 1.3	93.7 ± 1.1	94.0 ± 1.0	96.4 ± 0.9
	F1	87.2 ± 1.6	88.1 ± 1.4	91.4 ± 1.3	92.1 ± 1.1	94.1 ± 1.0
	PR AUC	91.1 ± 1.3	92.1 ± 1.1	95.5 ± 0.9	96.1 ± 0.8	97.9 ± 0.7
	MCC	86.4 ± 1.5	87.6 ± 1.3	90.8 ± 1.2	91.6 ± 1.0	93.6 ± 0.9
×4	Accuracy	89.0 ± 1.5	90.3 ± 1.3	91.8 ± 1.2	93.2 ± 1.1	96.1 ± 1.0
	F1	86.7 ± 1.7	88.3 ± 1.5	89.8 ± 1.4	91.0 ± 1.2	94.1 ± 1.1
	PR AUC	90.9 ± 1.4	92.3 ± 1.2	93.6 ± 1.0	94.9 ± 0.9	98.1 ± 0.8
	MCC	86.4 ± 1.6	87.2 ± 1.4	89.2 ± 1.3	91.0 ± 1.1	93.2 ± 1.0
×6	Accuracy	90.7 ± 1.3	92.4 ± 1.2	94.8 ± 1.0	95.0 ± 0.9	97.8 ± 0.8
	F1	88.4 ± 1.5	90.9 ± 1.3	92.9 ± 1.2	93.7 ± 1.0	96.8 ± 0.9
	PR AUC	93.6 ± 1.2	95.5 ± 1.0	96.9 ± 0.9	97.4 ± 0.8	99.4 ± 0.7
	MCC	88.2 ± 1.4	90.3 ± 1.2	92.5 ± 1.1	92.8 ± 1.0	95.3 ± 0.8
Tanzania						
×2	Accuracy	90.2 ± 1.4	91.8 ± 1.2	92.5 ± 1.1	94.2 ± 1.0	96.9 ± 0.9
	F1	88.0 ± 1.5	89.7 ± 1.3	90.1 ± 1.2	91.8 ± 1.1	94.8 ± 1.0
	PR AUC	91.8 ± 1.3	93.8 ± 1.1	94.7 ± 0.9	96.4 ± 0.8	98.8 ± 0.7
	MCC	87.4 ± 1.4	89.0 ± 1.2	90.4 ± 1.1	91.6 ± 1.0	94.4 ± 0.9
×4	Accuracy	87.6 ± 1.5	90.4 ± 1.4	92.2 ± 1.2	94.2 ± 1.0	97.5 ± 0.9
	F1	85.3 ± 1.6	88.3 ± 1.5	89.7 ± 1.3	92.1 ± 1.2	94.3 ± 1.0
	PR AUC	89.7 ± 1.4	92.8 ± 1.2	93.8 ± 1.0	96.2 ± 0.9	99.1 ± 0.8
	MCC	85.6 ± 1.5	88.1 ± 1.3	89.3 ± 1.2	92.0 ± 1.0	94.5 ± 0.9
×6	Accuracy	86.8 ± 1.6	89.3 ± 1.5	90.9 ± 1.3	93.4 ± 1.2	95.5 ± 1.1
	F1	84.9 ± 1.7	87.4 ± 1.5	89.0 ± 1.4	91.1 ± 1.3	93.2 ± 1.1
	PR AUC	88.9 ± 1.5	91.7 ± 1.3	93.1 ± 1.1	95.5 ± 1.0	97.6 ± 0.9
	MCC	83.9 ± 1.6	86.9 ± 1.4	88.1 ± 1.3	90.4 ± 1.1	92.7 ± 1.0
Plagues Maiz						
×2	Accuracy	86.1 ± 1.7	89.0 ± 1.5	90.9 ± 1.3	92.6 ± 1.2	95.4 ± 1.0
	F1	83.9 ± 1.8	86.6 ± 1.6	88.7 ± 1.4	90.7 ± 1.3	93.9 ± 1.1
	PR AUC	89.7 ± 1.6	91.5 ± 1.3	93.8 ± 1.2	95.8 ± 1.0	98.1 ± 0.9
	MCC	83.4 ± 1.7	86.0 ± 1.5	88.2 ± 1.3	90.1 ± 1.2	92.5 ± 1.0
×4	Accuracy	88.9 ± 1.5	90.1 ± 1.3	92.8 ± 1.2	94.5 ± 1.0	96.9 ± 0.9
	F1	86.5 ± 1.6	88.3 ± 1.4	90.5 ± 1.3	92.3 ± 1.1	95.8 ± 1.0
	PR AUC	92.3 ± 1.4	94.5 ± 1.2	96.1 ± 1.0	97.6 ± 0.9	99.4 ± 0.8
	MCC	86.3 ± 1.5	87.9 ± 1.3	90.1 ± 1.2	92.0 ± 1.0	94.2 ± 0.9
×6	Accuracy	87.1 ± 1.6	89.7 ± 1.4	91.6 ± 1.2	93.5 ± 1.0	95.7 ± 0.9
	F1	85.0 ± 1.7	87.4 ± 1.5	89.8 ± 1.3	91.4 ± 1.2	94.5 ± 1.0
	PR AUC	90.7 ± 1.5	93.5 ± 1.2	95.4 ± 1.0	96.6 ± 0.9	98.7 ± 0.8
	MCC	84.3 ± 1.6	86.8 ± 1.4	88.7 ± 1.2	90.5 ± 1.1	92.7 ± 0.9

Table 5. Performance metrics comparison across models and augmentation levels.

Dataverse. It leads on Tanzania as well, reporting 92.0 ± 1.2% at ×2. However, many CNNs decline sharply with increased augmentation. When comparing MaizeFormerX results to the best CNNs, the margins are significant: 97.8 ± 0.8% for MaizeFormerX on Dataverse ×6 versus EfficientNet-B1's 93.4 ± 1.0% (+4.4 points). On Tanzania ×2, MaizeFormerX scores 96.9 ± 0.9%, ahead of EfficientNet-B1's 92.0 ± 1.2% (+4.9 points). The largest gap is on Plagues Maiz ×4, where MaizeFormerX reaches 96.9 ± 0.9%, compared to EfficientNet-B1's 90.8 ± 1.2% (+6.1 points).

When comparing the best results from CNNs (see Table 6) to those of MaizeFormerX (refer to Table 5), the advantages of MaizeFormerX are significant across all datasets. For the Dataverse ×6 dataset, MaizeFormerX achieves an accuracy of 97.8 ± 0.8%, while the best CNN, EfficientNet-B1, reaches 93.4 ± 1.0%, resulting in a margin of approximately + 4.4% points. In the Tanzania ×2 dataset, MaizeFormerX records 96.9 ± 0.9%, compared to EfficientNet-B1's 92.0 ± 1.2%, which reflects a difference of about + 4.9 points. The largest gap is observed in the Plagues Maiz ×4 dataset, where MaizeFormerX reaches 96.9 ± 0.9%, while the best CNN baseline (EfficientNet-B1) reports 90.8 ± 1.2%. This indicates a margin of roughly + 6.1 points. Table 5 shows that MaizeFormerX consistently outperforms lightweight transformers, while Table 6 illustrates that it also surpasses the strongest mobile CNNs by a significant margin. This supports the assertion that MaizeFormerX provides a superior balance of accuracy and robustness under both moderate and strong augmentation settings.

Dataverse							
Augmentation	Metric	MobileNetV3	EffNet-B0	EffNet-B1	ShuffleNetV2	GhostNet	MixMobileNet
×2	Accuracy	91.6 ± 1.3	89.4 ± 1.8	92.7 ± 1.2	87.3 ± 2.1	90.8 ± 1.5	88.6 ± 2.0
	F1	88.8 ± 1.4	86.1 ± 1.9	89.9 ± 1.3	84.2 ± 2.2	87.7 ± 1.6	85.3 ± 2.1
	MCC	87.1 ± 1.4	83.7 ± 2.0	88.9 ± 1.3	81.0 ± 2.3	86.0 ± 1.7	82.8 ± 2.1
×4	Accuracy	90.2 ± 1.6	88.1 ± 2.0	92.9 ± 1.1	86.7 ± 2.3	91.3 ± 1.4	89.6 ± 1.8
	F1	87.6 ± 1.7	85.0 ± 2.1	88.7 ± 1.4	83.1 ± 2.4	86.9 ± 1.6	85.8 ± 1.9
	MCC	86.0 ± 1.6	82.7 ± 2.2	88.3 ± 1.4	80.2 ± 2.5	85.6 ± 1.7	83.9 ± 2.0
×6	Accuracy	92.1 ± 1.2	89.0 ± 1.9	93.4 ± 1.0	88.4 ± 2.2	91.7 ± 1.3	90.6 ± 1.7
	F1	89.7 ± 1.3	86.4 ± 2.0	91.1 ± 1.1	84.5 ± 2.3	88.9 ± 1.5	87.2 ± 1.9
	MCC	88.0 ± 1.3	84.0 ± 2.1	90.2 ± 1.2	82.1 ± 2.4	87.1 ± 1.6	84.8 ± 2.0
Tanzania							
Aug	Metric	MobileNetV3	EffNet-B0	EffNet-B1	ShuffleNetV2	GhostNet	MixMobileNet
×2	Accuracy	91.3 ± 1.3	88.0 ± 2.1	92.0 ± 1.2	86.4 ± 2.4	90.7 ± 1.5	89.1 ± 1.9
	F1	89.0 ± 1.4	85.6 ± 2.2	90.1 ± 1.3	83.2 ± 2.6	87.6 ± 1.7	86.8 ± 2.0
	MCC	87.2 ± 1.4	83.1 ± 2.3	88.5 ± 1.3	80.6 ± 2.5	85.9 ± 1.8	84.2 ± 2.1
×4	Accuracy	90.1 ± 1.4	84.7 ± 2.4	91.9 ± 1.1	82.3 ± 2.7	88.8 ± 1.7	86.1 ± 2.2
	F1	86.9 ± 1.6	81.2 ± 2.6	88.9 ± 1.3	79.0 ± 2.9	85.7 ± 1.9	82.6 ± 2.4
	MCC	85.0 ± 1.5	79.1 ± 2.6	87.4 ± 1.3	76.8 ± 2.8	83.9 ± 2.0	80.6 ± 2.5
×6	Accuracy	90.9 ± 1.3	83.2 ± 2.6	91.4 ± 1.2	81.1 ± 2.9	87.6 ± 2.0	84.5 ± 2.4
	F1	87.8 ± 1.5	79.8 ± 2.8	88.2 ± 1.3	77.2 ± 3.0	84.5 ± 2.2	81.1 ± 2.6
	MCC	85.9 ± 1.5	77.6 ± 2.7	86.1 ± 1.4	75.1 ± 3.0	82.7 ± 2.3	79.3 ± 2.6
Plagues Maiz							
Aug	Metric	MobileNetV3	EffNet-B0	EffNet-B1	ShuffleNetV2	GhostNet	MixMobileNet
×2	Accuracy	87.8 ± 1.6	81.9 ± 2.6	88.6 ± 1.4	79.3 ± 2.9	86.4 ± 1.7	83.7 ± 2.3
	F1	84.6 ± 1.8	78.1 ± 2.8	85.2 ± 1.6	75.0 ± 3.1	82.7 ± 1.9	79.9 ± 2.6
	MCC	83.0 ± 1.7	76.0 ± 2.7	83.9 ± 1.6	72.8 ± 3.0	81.1 ± 2.0	78.2 ± 2.6
×4	Accuracy	89.1 ± 1.4	86.0 ± 2.1	90.8 ± 1.2	82.6 ± 2.8	88.7 ± 1.5	84.9 ± 2.6
	F1	86.2 ± 1.6	82.3 ± 2.3	88.6 ± 1.3	78.7 ± 3.0	85.4 ± 1.7	81.1 ± 2.8
	MCC	84.6 ± 1.5	80.4 ± 2.2	87.2 ± 1.3	76.1 ± 2.9	83.8 ± 1.8	79.2 ± 2.7
×6	Accuracy	90.7 ± 1.3	82.7 ± 2.7	91.2 ± 1.1	81.0 ± 3.1	88.9 ± 1.6	84.1 ± 2.9
	F1	87.4 ± 1.4	79.2 ± 2.9	89.1 ± 1.2	77.0 ± 3.2	85.8 ± 1.8	80.5 ± 3.0
	MCC	85.5 ± 1.4	77.0 ± 2.8	88.0 ± 1.2	74.7 ± 3.1	84.0 ± 1.9	78.3 ± 2.9

Table 6. Performance comparison of lightweight CNN baselines on the Dataverse, Tanzania, and Plagues Maiz datasets across models and augmentation levels.

Table 7 shows that MaizeFormerX outperforms other models without any augmentation, indicating strong intrinsic representation quality. On the Dataverse dataset, it achieves 94.0 ± 1.0% accuracy, surpassing the best lightweight transformer, TinyViT, by 1.9 points (92.1 ± 1.3%). This is also reflected in confidence-robust metrics, with MaizeFormerX achieving a PR-AUC of 96.1 ± 0.9, significantly higher than Swin Transformer V2's 93.6 ± 1.0, demonstrating better ranking of positives and negatives. The trend continues on the Tanzania dataset, where MaizeFormerX records 92.3 ± 1.2% accuracy compared to TinyViT's 90.1 ± 1.5%. It also shows improved balanced discrimination with an MCC of 90.2 ± 1.3, outperforming Swin V2's 86.7 ± 1.4, indicating fewer class confusions. On the Plagues Maiz dataset, MaizeFormerX maintains its lead with 92.9 ± 1.1% accuracy, ahead of TinyViT (91.2 ± 1.3%) and Swin V2 (90.6 ± 1.2%), confirming its consistent performance across different datasets.

Figure 12 shows that MaizeFormerX achieves an impressive accuracy–robustness profile under various acquisition conditions while being lightweight. Panels A–B indicate that it has the highest best-augmentation MCC across all datasets: 95.3% on Dataverse, 94.4% on Tanzania, and 94.2% on Plagues Maiz, outperforming the strongest ViT baselines (e.g., 92.8% on Dataverse, 91.6% on Tanzania, and 92.0% on Plagues Maiz) and demonstrating better performance than CNN baselines, which struggle under domain variability. Panel C underscores the differing impacts of augmentation on robustness across datasets. The most significant improvement, from 90.3% to 94.2% (+3.9 MCC), occurs in the challenging Plagues Maiz domain, reflecting enhanced resilience to class imbalance. Dataverse also sees a notable gain from 91.6% to 95.3% (+3.7), indicating better lesion–background separation, even in cleaner conditions. For Tanzania, the improvement from 90.2% to 94.4% (+4.2) is smaller but peaks at moderate augmentation, suggesting that overly strong augmentation may lead to feature drift. Panel D highlights MaizeFormerX's superiority against top baselines at the best augmentation. It achieves +2.5 MCC on Dataverse compared to the strongest ViT (95.3 vs. 92.8), +2.8 on Tanzania (94.4 vs. 91.6), and +2.2 on Plagues Maiz (94.2 vs. 92.0). The margin over CNN baselines is consistent, confirming that

Metric	MobileViT	EfficientFormer	TinyViT	Swin Transformer V2	MaizeFormerX
Dataverse					
Accuracy	89.5 ± 1.4	90.9 ± 1.5	92.1 ± 1.3	91.6 ± 1.2	94.0 ± 1.0
F1	87.7 ± 1.5	89.2 ± 1.4	90.8 ± 1.2	91.0 ± 1.1	93.4 ± 1.0
PR AUC	91.0 ± 1.3	91.9 ± 1.3	94.0 ± 1.1	93.6 ± 1.0	96.1 ± 0.9
MCC	87.0 ± 1.5	88.8 ± 1.4	90.2 ± 1.3	89.3 ± 1.2	91.6 ± 1.0
Tanzania					
Accuracy	86.9 ± 1.8	88.7 ± 1.6	90.1 ± 1.5	89.4 ± 1.4	92.3 ± 1.2
F1	85.2 ± 1.7	87.0 ± 1.5	88.7 ± 1.4	88.1 ± 1.4	91.0 ± 1.2
PR AUC	88.2 ± 1.6	89.7 ± 1.5	91.9 ± 1.3	91.0 ± 1.3	94.2 ± 1.1
MCC	84.5 ± 1.7	86.4 ± 1.6	88.0 ± 1.5	86.7 ± 1.4	90.2 ± 1.3
Plagues Maiz					
Accuracy	88.1 ± 1.6	89.5 ± 1.5	91.2 ± 1.3	90.6 ± 1.2	92.9 ± 1.1
F1	86.1 ± 1.6	88.2 ± 1.5	89.7 ± 1.3	89.8 ± 1.3	91.8 ± 1.1
PR AUC	89.7 ± 1.4	91.3 ± 1.2	92.8 ± 1.2	92.0 ± 1.1	95.4 ± 1.0
MCC	85.6 ± 1.5	87.2 ± 1.4	89.1 ± 1.3	88.4 ± 1.2	90.3 ± 1.1

Table 7. Performance metrics comparison across models on non-augmented datasets.

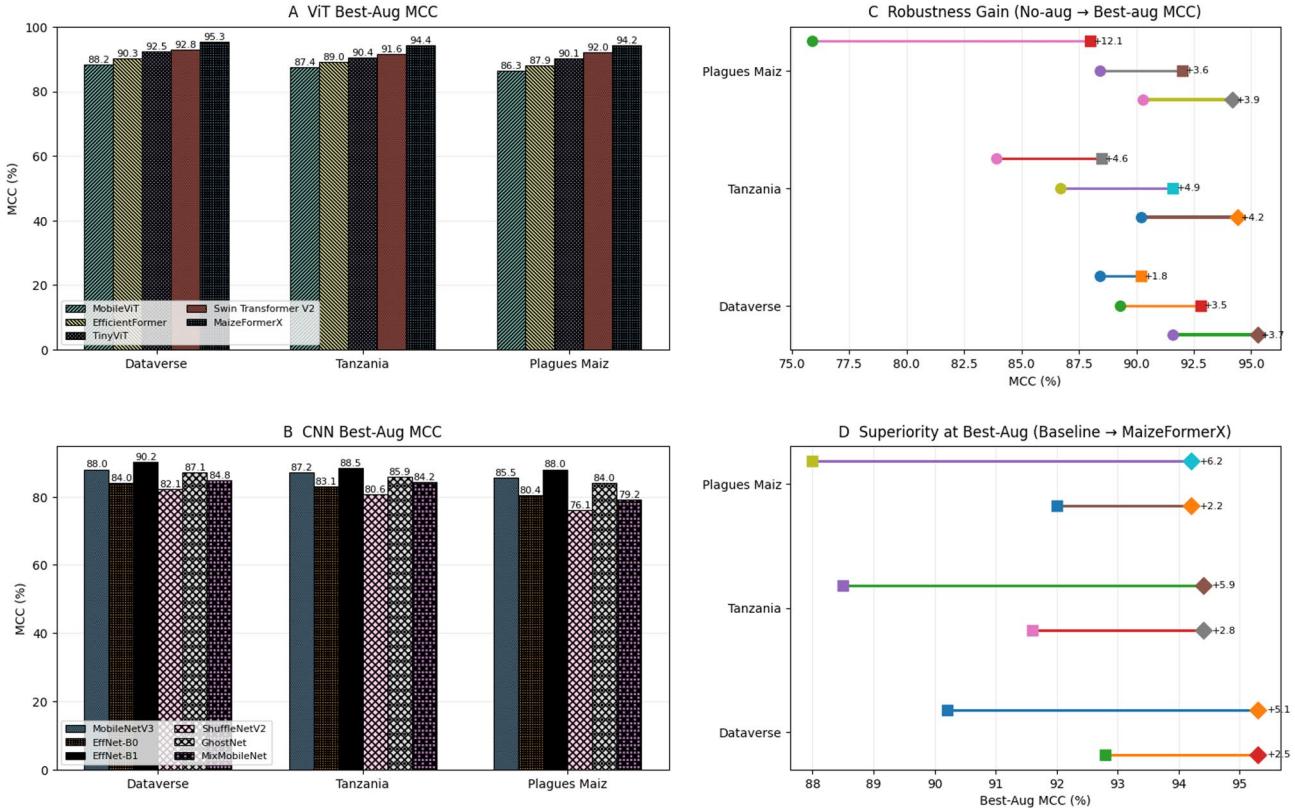


Fig. 12. Consolidated performance visualization. (**A**) report the best-augmentation MCC for Vision Transformer baselines and MaizeFormerX across Dataverse, Tanzania, and Plagues Maiz. (**B**) the best-augmentation MCC for lightweight CNN baselines across the same datasets. (**C**) quantify robustness gain from augmentation by showing no-augmentation → best-augmentation MCC for the strongest CNN baseline, the strongest ViT baseline, and MaizeFormerX. (**D**) quantify superiority at best augmentation by comparing the best CNN baseline → MaizeFormerX and best ViT baseline → MaizeFormerX for each dataset (ΔMCC annotated).

the proposed cross-scale attention fusion enhances representation quality beyond what lightweight CNNs or compact ViTs can achieve alone.

Table 8 summarizes mobile CNN performance without augmentation, highlighting two key trends: (i) performance ceilings are generally lower than MaizeFormerX, and (ii) the top-performing CNN varies by dataset.

Metric	MobileNetV3	EfficientNet-B0	EfficientNet-B1	ShuffleNetV2	GhostNet	MixMobileNet
Dataverse						
Accuracy	91.5 ± 1.3	88.7 ± 1.9	91.2 ± 1.4	87.9 ± 2.2	90.8 ± 1.6	89.6 ± 2.0
F1	90.4 ± 1.3	86.2 ± 2.0	89.5 ± 1.5	84.7 ± 2.4	88.9 ± 1.7	87.3 ± 2.1
PR AUC	93.3 ± 1.1	89.4 ± 1.9	92.8 ± 1.2	88.1 ± 2.3	92.0 ± 1.4	90.7 ± 2.0
MCC	89.0 ± 1.3	83.9 ± 2.1	88.4 ± 1.4	81.8 ± 2.5	87.1 ± 1.6	84.6 ± 2.2
Tanzania						
Accuracy	89.1 ± 1.5	84.9 ± 2.5	87.6 ± 1.8	82.6 ± 2.8	89.3 ± 1.4	86.0 ± 2.3
F1	87.8 ± 1.6	80.9 ± 2.7	85.7 ± 2.0	78.5 ± 3.0	88.0 ± 1.4	83.3 ± 2.5
PR AUC	90.6 ± 1.3	85.8 ± 2.4	88.9 ± 1.8	83.7 ± 2.9	90.9 ± 1.2	87.1 ± 2.6
MCC	86.3 ± 1.6	78.8 ± 2.8	83.9 ± 2.0	76.4 ± 3.1	86.6 ± 1.4	80.6 ± 2.7
Plagues Maiz						
Accuracy	90.2 ± 1.4	85.1 ± 2.6	82.9 ± 3.0	90.5 ± 1.2	88.7 ± 1.8	86.4 ± 2.4
F1	89.2 ± 1.5	81.7 ± 2.9	78.9 ± 3.2	89.6 ± 1.3	86.4 ± 2.1	83.8 ± 2.7
PR AUC	91.4 ± 1.3	87.0 ± 2.5	84.6 ± 3.0	91.9 ± 1.1	90.1 ± 1.7	88.2 ± 2.6
MCC	87.6 ± 1.5	79.6 ± 2.8	75.9 ± 3.1	88.2 ± 1.2	84.3 ± 2.0	80.8 ± 2.7

Table 8. Performance comparison of lightweight CNN baselines on the Dataverse, Tanzania, and Plagues Maiz datasets without augmentation.

On Dataverse, MobileNetV3 achieves the highest accuracy at 91.5 ± 1.3%, followed closely by EfficientNet-B1 at 91.2 ± 1.4%. Some models, like ShuffleNetV2, drop significantly to 87.9 ± 2.2%, indicating sensitivity to limited training diversity. In Tanzania, GhostNet is the best CNN at 89.3 ± 1.4%, with MobileNetV3 at 89.1 ± 1.5%, but both are below the MaizeFormerX ceiling. For Plagues Maiz, ShuffleNetV2 performs best at 90.5 ± 1.2%, slightly ahead of MobileNetV3 at 90.2 ± 1.4%, illustrating that CNN performance is dataset-dependent without augmentation.

On the other hand, Tables 7 and 8 demonstrate that MaizeFormerX outperforms both lightweight transformer and mobile CNN baselines in the non-augmented regime. It achieves significant accuracy improvements over the best CNNs: 94.0 vs. 91.5 (+2.5) on Dataverse, 92.3 vs. 89.3 (+3.0) on Tanzania, and 92.9 vs. 90.5 (+2.4) on Plagues Maiz. Additionally, it significantly enhances balanced-quality measures, with MCC improvements of about + 2.6 on Dataverse (91.6 vs. 89.0) and + 3.6 on Tanzania (90.2 vs. 86.6), indicating better class separation beyond just minor top-1 correctness gains.

Figure 13 shows that augmentation impacts datasets differently. In Dataverse, stronger augmentation improves both separability and stability: the macro-average F1 increases from 97.97% at ×2 to 98.60% at ×6, while macro MCC rises from 95.75% to 96.26%, and macro PR-AUC increases from 99.16% to 99.36% (see Table 9). Class-wise improvements include Healthy F1 rising from 97.72 to 98.65 (+0.93) and MSV F1 from 97.90 to 98.60 (+0.70). The ×6 cluster is compact with an F1 range of 98.55–98.65, indicating reduced class-to-class variability. Notably, MACC dips slightly at ×4 (approximately 95.26) before peaking at ×6, highlighting that the greatest benefits occur with higher augmentation (see Table 9).

On the other hand, Tanzania exhibits over-augmentation sensitivity: macro F1 decreases from 97.83% (×2) to 97.12% (×6) and macro MCC drops from 95.43% to 93.91% (≈ -1.52), with PR-AUC declining 99.05%→98.74%, implying that aggressive perturbations reduce both discrimination and ranking confidence (Table 9). For Plagues Maiz (Panel C), class-level behavior becomes less stable at higher augmentation: macro F1 falls from 94.43% (×2) to 90.52% (×6), with large degradations in sensitive categories such as Cercospora zeae-maydis (95.66→88.43, −7.23) and Exserohilum turcicum (95.36→88.32, −7.04), whereas Virus (93.00→92.25, −0.75) and Colletotrichum graminicola (94.77→93.70, −1.07) remain relatively robust (Table 10). Panel D summarizes these shifts: Plagues shows a downward and wider per-class F1 distribution (mean 94.43→90.52; σ 1.11→1.66; min 92.74→87.44), supporting the conclusion that ×6 is beneficial for Dataverse, while ×2 (or ×4) is safer for Tanzania and Plagues Maiz (Tables 9 and 10).

Cross-dataset generalization

Table 8 shows that MaizeFormerX consistently outperforms all models in cross-dataset generalization, leading by 3–6% points over the best baseline, Swin Transformer V2. While data augmentation boosts cross-domain accuracy, its benefits level off beyond 2× to 6× augmentation, adding only an average of 1–2% points. The biggest improvement occurs when training on Tanzania and testing on the difficult Plagues Maiz domain, where MaizeFormerX achieves 86.8%. In easier transfers, such as Dataverse to Tanzania, accuracy stabilizes, indicating diminishing returns with excessive augmentation. Domain shift is a major challenge, as using Plagues Maiz for testing consistently lowers performance due to differences in imaging conditions and class imbalance. For example, models trained on Tanzania lose 5–6% points when evaluated on Plagues Maiz. Swin Transformer V2 remains the closest competitor, consistently coming in second and benefiting slightly from augmentation. TinyViT surpasses EfficientFormer and MobileViT, showcasing the advantages of a compact yet deep transformer design, while MobileViT performs poorly under low augmentation, reaching only 68.9% accuracy when transferring from Dataverse to Plagues Maiz, highlighting its architectural limitations.

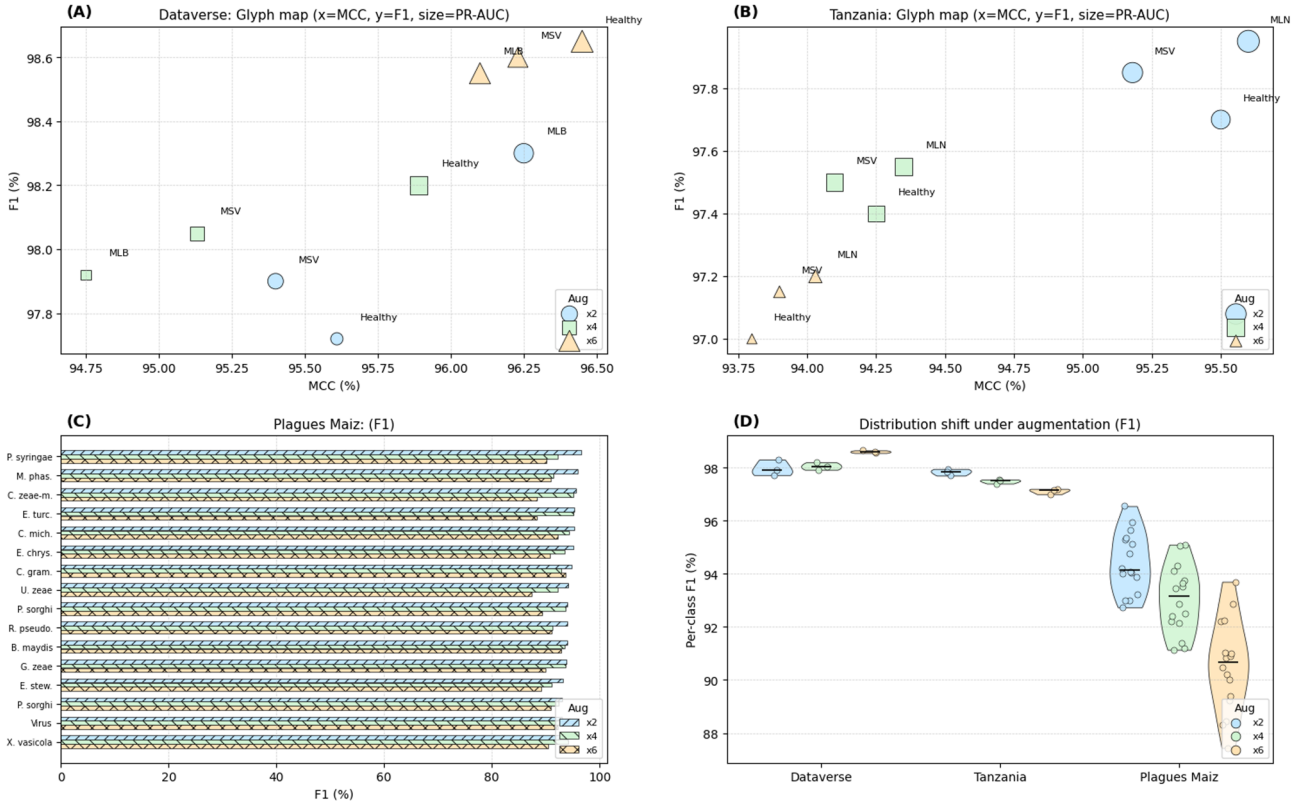


Fig. 13. Augmentation sensitivity for Dataverse and Tanzania : x-axis = MCC, y-axis = F1, marker size = PR-AUC, indicating ×2/×4/×6. (C) per-class F1 on Plagues Maiz across ×2/×4/×6. (D) per-class F1 distributions for each dataset across augmentation strengths.

Augmentation	Class	Accuracy (%)	MCC (%)	PR AUC (%)	F1 Score (%)
Dataverse					
×2	Healthy	97.58	95.61	99.05	97.72
	MLB	98.12	96.25	99.29	98.3
	MSV	97.65	95.4	99.15	97.9
×4	Healthy	98.04	95.89	99.22	98.2
	MLB	97.55	94.75	99	97.92
	MSV	97.89	95.13	99.1	98.05
×6	Healthy	98.48	96.45	99.4	98.65
	MLB	98.36	96.1	99.36	98.55
	MSV	98.25	96.23	99.32	98.6
Tanzania					
×2	Healthy	97.3	95.5	98.98	97.7
	MLN	97.65	95.6	99.12	97.95
	MSV	97.48	95.18	99.04	97.85
×4	Healthy	96.92	94.25	98.87	97.4
	MLN	97.08	94.35	98.95	97.55
	MSV	96.98	94.1	98.9	97.5
×6	Healthy	96.55	93.8	98.7	97
	MLN	96.78	94.03	98.78	97.2
	MSV	96.7	93.9	98.74	97.15

Table 9. Classification report of MaizeFormerX on ×2, ×4, and ×6 augmented training data for the Dataverse and Tanzania datasets.

Augmentation	Class	Accuracy (%)	MCC (%)	PR AUC (%)	F1 Score (%)
×2	Erwinia chrysanthemi	94.85	92.98	96.4	95.12
	Erwinia stewartii	93.21	90.45	94.55	93.22
	Clavibacter michiganensis	95.43	92.84	96.71	95.3
	Pseudomonas syringae	96.1	93.75	98.05	96.55
	Rubrilineans pseudomonas	93.54	90.8	95.13	94.05
	Xanthomonas vasicola	93.09	90.55	94.02	92.74
	Bipolaris maydis	94.22	91.98	96.08	94
	Cercospora zeae-maydis	95.7	93.1	97.24	95.66
	Colletotrichum graminicola	94.31	91.8	96.83	94.77
	Gibberella zeae	93.77	91.15	94.63	93.88
	Macrophomina phaseolina	96.12	93.34	97.11	95.96
	Peronosclerospora sorghi	94.3	91.82	95.42	94.08
	Puccinia sorghi	93.6	91.27	94.83	93.01
	Exserohilum turcicum	95.48	92.76	96.95	95.36
	Ustilago zeae	94.15	91.45	95.37	94.2
	Virus	93.01	90.62	94.42	93
×4	Erwinia chrysanthemi	92.85	90.26	95.22	93.51
	Erwinia stewartii	91.68	88.7	93.75	91.14
	Clavibacter michiganensis	93.7	91	96.3	94.3
	Pseudomonas syringae	92.14	89.82	94.45	92.2
	Rubrilineans pseudomonas	91.9	89.44	93.66	91.2
	Xanthomonas vasicola	94.35	91.7	96.12	94.12
	Bipolaris maydis	93.3	90.53	95.6	93.45
	Cercospora zeae-maydis	94.7	91.93	97.03	95.1
	Colletotrichum graminicola	92	89.64	94.97	92.85
	Gibberella zeae	93.65	90.9	96.13	93.75
	Macrophomina phaseolina	91.32	88.78	93.7	91.4
	Peronosclerospora sorghi	93.14	90.44	96.06	93.65
	Puccinia sorghi	92.72	90.16	95.31	92.4
	Exserohilum turcicum	94.82	92.18	96.72	95.06
	Ustilago zeae	91.89	89.44	94.12	92.15
	Virus	92.31	89.87	95	92.5
×6	Erwinia chrysanthemi	91.02	88.33	92.67	90.85
	Erwinia stewartii	89.55	86.94	91.03	89.22
	Clavibacter michiganensis	92.66	90.02	94.12	92.21
	Pseudomonas syringae	90.73	88.12	91.85	90.2
	Rubrilineans pseudomonas	91.55	88.66	93.14	91.02
	Xanthomonas vasicola	90.92	88.05	92.41	90.48
	Bipolaris maydis	93.03	90.55	95.2	92.85
	Cercospora zeae-maydis	89.67	87	91.35	88.43
	Colletotrichum graminicola	93.25	90.36	95.82	93.7
	Gibberella zeae	90.28	87.62	92.93	90.03
	Macrophomina phaseolina	91.44	88.99	93.31	90.9
	Peronosclerospora sorghi	89.91	87.06	91.13	89.4
	Puccinia sorghi	91.12	88.52	93.47	91
	Exserohilum turcicum	88.8	86.06	90.83	88.32
	Ustilago zeae	87.9	85.05	90.21	87.44
	Virus	92.08	89.67	94.78	92.25

Table 10. Classification report of MaizeFormerX on ×2, ×4, and ×6 augmented training data for the Plagues Maiz dataset.

Component analysis

Component analysis in Table 9 shows that the full MaizeFormerX model consistently outperforms all ablated configurations, highlighting the importance of each architectural component. The removal of the cross-scale attention module leads to the largest performance drop, decreasing accuracy by 2–4% points and MCC across all datasets (e.g., from 97.5 to 95.1% accuracy and from 96.3 to 92.4% MCC on the Dataverse dataset at ×6

augmentation). The absence of transformer blocks also results in a 1–2% point accuracy reduction. Multi-scale fusion provides incremental benefits; its removal causes a smaller accuracy drop of about 0.7–1.2% points (e.g., from 95.8 to 91.1% on Plagues Maiz at ×6), helping fine-grained classification by integrating features across spatial levels. While different augmentation levels impact overall accuracy, they do not change the relative importance of these components: Dataverse peaks at ×6, Tanzania at ×2, and Plagues Maiz at ×4, illustrating dataset-specific trade-offs between variability and overfitting. As illustrated in Fig. 14, the full MaizeFormerX consistently outperforms its ablated variants across all datasets and augmentation levels (Tables 11, 12).

Statistical significance test

Table 10 summarizes the Wilcoxon signed-rank test results, clearly demonstrating that MaizeFormerX surpasses existing models. The consistently low (p)-values across various models and datasets indicate significant advantages, with most values below 0.01, well under the 0.05 threshold. This confirms the effectiveness of MaizeFormerX's architectural enhancements and its capability to handle different augmentation levels and data diversity. When compared with MobileViT, the (p)-values range from 0.0016 to 0.0033, showing it is the model MaizeFormerX outperforms the most. Comparisons with EfficientFormer and TinyViT yield (p)-values between 0.0035 and 0.0069. Even the strong Swin Transformer V2 displays (p)-values between 0.0075 and 0.0095, indicating that MaizeFormerX maintains a marginal yet significant edge. Notably, the lowest (p)-values often occur at the $(\times 4)$ augmentation level, suggesting this may best showcase MaizeFormerX's strengths. However, the low (p)-values are consistent across $(\times 2)$, $(\times 4)$, and $(\times 6)$ levels, indicating its performance advantage is robust. MaizeFormerX shows statistical superiority across all three datasets. A limitation of this analysis is the absence of effect size metrics, like Cohen's (d) or Cliff's Delta, which would help assess the practical significance of the results. While the (p)-values indicate statistical significance, effect sizes would provide better insights into real-world relevance (Table 13).

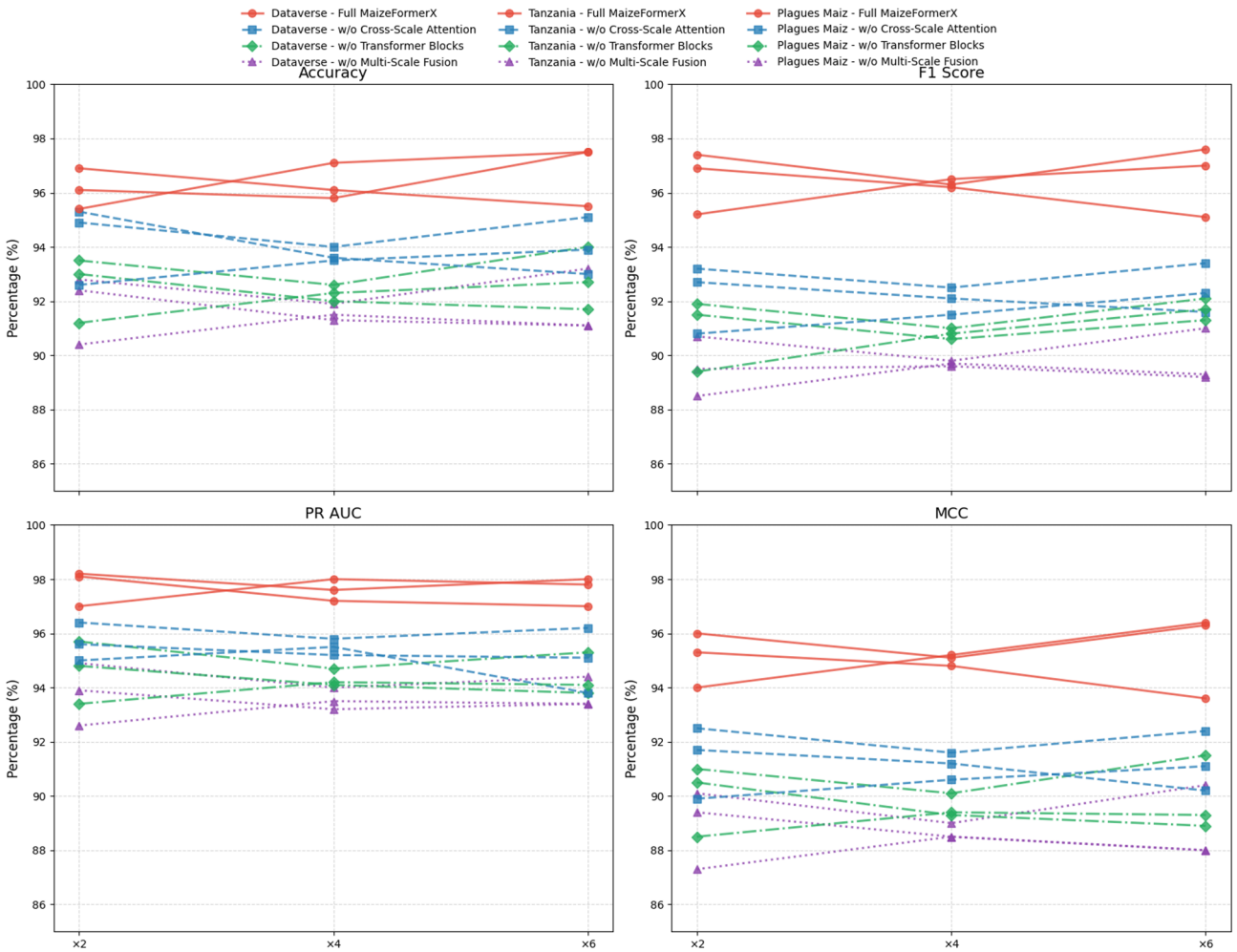


Fig. 14. Performance comparison of MaizeFormerX and its ablated variants across datasets and augmentation levels.

Train → Test	MobileViT	EfficientFormer	TinyViT	Swin Transformer V2	MaizeFormerX
Dataverse (×2)					
→ Tanzania	72.6 ± 1.8	76.4 ± 1.6	81.7 ± 1.3	84.2 ± 1.1	88.9 ± 1.0
→ Plagues Maiz	68.9 ± 2.0	74.1 ± 1.7	77.3 ± 1.4	81.5 ± 1.2	86.7 ± 1.1
Dataverse (×4)					
→ Tanzania	70.2 ± 1.7	75.8 ± 1.5	79.1 ± 1.3	82.9 ± 1.1	87.3 ± 1.0
→ Plagues Maiz	69.6 ± 1.8	73.5 ± 1.6	78.4 ± 1.3	80.8 ± 1.2	85.9 ± 1.1
Dataverse (×6)					
→ Tanzania	71.5 ± 1.6	76.1 ± 1.4	80.3 ± 1.2	83.4 ± 1.1	88.0 ± 1.0
→ Plagues Maiz	70.8 ± 1.7	74.6 ± 1.5	79.7 ± 1.3	82.0 ± 1.2	86.6 ± 1.1
Tanzania (×2)					
→ Dataverse	69.5 ± 1.9	74.7 ± 1.7	78.8 ± 1.4	81.2 ± 1.2	86.1 ± 1.1
→ Plagues Maiz	67.8 ± 2.0	72.6 ± 1.8	76.9 ± 1.5	79.1 ± 1.3	84.5 ± 1.2
Tanzania (×4)					
→ Dataverse	66.3 ± 2.1	71.2 ± 1.9	76.5 ± 1.6	79.8 ± 1.3	84.9 ± 1.2
→ Plagues Maiz	64.7 ± 2.2	70.1 ± 2.0	74.9 ± 1.8	77.3 ± 1.5	82.7 ± 1.4
Tanzania (×6)					
→ Dataverse	70.1 ± 1.8	75.4 ± 1.6	80.0 ± 1.4	83.1 ± 1.2	87.7 ± 1.1
→ Plagues Maiz	68.3 ± 1.9	73.7 ± 1.7	77.8 ± 1.5	81.0 ± 1.3	86.8 ± 1.2
Plagues Maiz (×2)					
→ Dataverse	68.7 ± 1.9	73.1 ± 1.6	77.2 ± 1.4	80.3 ± 1.2	85.1 ± 1.1
→ Tanzania	70.5 ± 1.8	75.9 ± 1.6	79.8 ± 1.3	82.7 ± 1.1	88.0 ± 1.0
Plagues Maiz (×4)					
→ Dataverse	71.4 ± 1.6	76.2 ± 1.4	80.5 ± 1.2	83.8 ± 1.0	87.2 ± 0.9
→ Tanzania	72.2 ± 1.5	77.4 ± 1.3	81.3 ± 1.1	84.9 ± 1.0	86.1 ± 0.9
Plagues Maiz (×6)					
→ Dataverse	67.9 ± 2.0	72.3 ± 1.8	76.1 ± 1.6	79.5 ± 1.4	84.4 ± 1.3
→ Tanzania	69.2 ± 1.9	74.4 ± 1.7	78.7 ± 1.5	81.9 ± 1.2	86.6 ± 1.1

Table 11. Cross-dataset generalization performance (at $\times 2$, $\times 4$, $\times 6$ augmentations): models trained on one dataset and tested on others.

Model complexity analysis

Table 14 compares the model complexity and inference efficiency of MaizeFormerX with lightweight CNN and compact ViT baselines across four key metrics: Params, GFLOPs, CPU latency, and GPU latency. MaizeFormerX employs 9.6 M parameters and 1.05 GFLOPs, achieving inference times of 16.4 ms on CPU (i9-13900 K) and 1.42 ms on GPU (RTX 4090). These measurements are significantly lower than those of heavyweight transformer models like Swin Transformer V2 (Tiny), which has 29.4 M params and much longer inference times. Compared to other lightweight ViTs, MaizeFormerX maintains a competitive edge. TinyViT has a similar parameter count (10.4 M) but requires more compute (2.12 GFLOPs) and has slower CPU inference (21.8 ms). MobileViT is lighter and faster but sacrifices some predictive accuracy in benchmarks. CNN models, such as ShuffleNetV2 and GhostNet, remain more efficient overall, highlighting that MaizeFormerX is better positioned as a lightweight transformer rather than being the lightest option among CNNs.

Figure 15 illustrates these efficiency trade-offs, showing that CNN models cover larger areas due to lower FLOPs and latency, while heavyweight transformers are constrained by higher compute requirements. MaizeFormerX achieves a balance, being more efficient than Swin-Tiny and heavier ViTs while being moderately heavier than the smallest mobile CNNs, making it suitable for smart-agriculture applications.

Model validation

Figure 16 illustrates the training and validation performance of the MaizeFormerX model over 30 epochs. Both loss curves decline steadily, indicating effective optimization on the study dataset. The close alignment between the training and validation loss suggests no clear evidence of severe overfitting in the evaluated splits. By around epoch 20, both training and validation accuracy approach 96%, with validation accuracy occasionally exceeding training accuracy, likely due to regularization effects, such as dropout. Recall improves consistently to above 90%, indicating better sensitivity to positive cases, while precision increases to nearly 98.2%, reflecting a reduced tendency for false positives in the current evaluation setting.

The learning curves presented in Fig. 17 suggest that the model is well optimized for the Tanzania dataset and achieves strong classification performance under the reported experimental conditions. Both training and validation loss decrease steadily to below 0.5 after 30 epochs, and their minimal divergence indicates stable learning behavior within the selected data splits. Accuracy improves rapidly during the early epochs and then stabilizes above 98%, with validation accuracy slightly higher than training accuracy, possibly due to

Augmentation	Configuration	Accuracy (%)	F1 Score (%)	PR AUC (%)	MCC (%)
Dataverse					
×2	Full MaizeFormerX	96.1 ± 0.8	97.4 ± 0.7	98.2 ± 0.6	96.0 ± 0.9
	w/o Cross-Scale Attention	94.9 ± 1.1	93.2 ± 1.5	96.4 ± 0.9	92.5 ± 1.2
	w/o Transformer Blocks	93.5 ± 1.3	91.9 ± 1.7	95.7 ± 1.0	91.0 ± 1.5
	w/o Multi-Scale Fusion	92.8 ± 1.6	90.7 ± 1.8	94.9 ± 1.1	90.1 ± 1.7
×4	Full MaizeFormerX	95.8 ± 1.0	96.3 ± 1.2	97.6 ± 0.8	95.1 ± 1.1
	w/o Cross-Scale Attention	94.0 ± 1.4	92.5 ± 1.5	95.8 ± 1.0	91.6 ± 1.4
	w/o Transformer Blocks	92.6 ± 1.5	91.0 ± 1.8	94.7 ± 1.2	90.1 ± 1.7
	w/o Multi-Scale Fusion	91.9 ± 1.7	89.8 ± 1.9	94.0 ± 1.3	89.0 ± 1.6
×6	Full MaizeFormerX	97.5 ± 0.6	97.6 ± 0.7	98.6 ± 0.5	96.3 ± 0.7
	w/o Cross-Scale Attention	95.1 ± 1.1	93.4 ± 1.3	96.2 ± 0.9	92.4 ± 1.2
	w/o Transformer Blocks	94.0 ± 1.4	92.1 ± 1.6	95.3 ± 1.0	91.5 ± 1.4
	w/o Multi-Scale Fusion	93.2 ± 1.5	91.0 ± 1.7	94.4 ± 1.2	90.4 ± 1.5
Tanzania					
×2	Full MaizeFormerX	96.9 ± 0.8	96.9 ± 0.9	98.1 ± 0.6	95.3 ± 0.9
	w/o Cross-Scale Attention	95.3 ± 1.3	92.7 ± 1.6	95.6 ± 0.9	91.7 ± 1.4
	w/o Transformer Blocks	93.0 ± 1.4	91.5 ± 1.7	94.8 ± 1.1	90.5 ± 1.5
	w/o Multi-Scale Fusion	92.4 ± 1.6	90.6 ± 1.8	94.1 ± 1.2	89.4 ± 1.6
×4	Full MaizeFormerX	96.1 ± 0.4	96.2 ± 1.1	97.4 ± 0.7	94.8 ± 1.0
	w/o Cross-Scale Attention	93.6 ± 1.4	92.1 ± 1.5	95.2 ± 1.0	91.2 ± 1.3
	w/o Transformer Blocks	92.0 ± 1.6	90.4 ± 1.8	94.0 ± 1.2	89.3 ± 1.7
	w/o Multi-Scale Fusion	91.3 ± 1.7	89.6 ± 1.9	93.2 ± 1.4	88.5 ± 1.8
×6	Full MaizeFormerX	95.5 ± 1.1	95.4 ± 1.3	97.0 ± 0.8	93.6 ± 1.1
	w/o Cross-Scale Attention	93.0 ± 1.5	91.2 ± 1.6	95.1 ± 1.0	90.2 ± 1.5
	w/o Transformer Blocks	91.7 ± 1.7	90.0 ± 1.8	93.7 ± 1.3	88.9 ± 1.7
	w/o Multi-Scale Fusion	91.1 ± 1.8	89.2 ± 2.0	93.0 ± 1.4	88.0 ± 1.8
Plagues Maiz					
×2	Full MaizeFormerX	95.4 ± 1.2	95.2 ± 1.4	97.0 ± 0.9	94.0 ± 1.1
	w/o Cross-Scale Attention	92.6 ± 1.5	90.8 ± 1.8	95.0 ± 1.2	89.9 ± 1.6
	w/o Transformer Blocks	91.2 ± 1.7	89.4 ± 1.9	93.4 ± 1.4	88.5 ± 1.7
	w/o Multi-Scale Fusion	90.4 ± 1.9	88.5 ± 2.0	92.6 ± 1.5	87.3 ± 1.9
×4	Full MaizeFormerX	97.1 ± 0.9	96.5 ± 1.1	98.0 ± 0.7	95.2 ± 1.0
	w/o Cross-Scale Attention	93.5 ± 1.4	91.9 ± 1.6	95.5 ± 1.0	90.6 ± 1.5
	w/o Transformer Blocks	92.3 ± 1.6	90.8 ± 1.8	94.2 ± 1.2	89.4 ± 1.7
	w/o Multi-Scale Fusion	91.5 ± 1.7	89.9 ± 1.9	93.5 ± 1.3	88.6 ± 1.8
×6	Full MaizeFormerX	95.8 ± 1.0	95.9 ± 1.2	97.6 ± 0.8	94.1 ± 1.1
	w/o Cross-Scale Attention	93.2 ± 1.5	91.5 ± 1.7	95.0 ± 1.1	90.4 ± 1.6
	w/o Transformer Blocks	91.9 ± 1.6	90.3 ± 1.8	93.8 ± 1.3	89.2 ± 1.7
	w/o Multi-Scale Fusion	91.1 ± 1.7	89.4 ± 1.9	93.0 ± 1.4	88.0 ± 1.8

Table 12. Comprehensive components study of MaizeFormerX across all datasets and augmentation levels.

regularization. Recall exceeds 90%, and precision surpasses 95%, indicating a favorable balance between false negatives and false positives.

Figure 18 shows the training and validation performance of MaizeFormerX on the Plagues Maiz dataset. The loss curves decrease from above 2.0 to approximately 0.4, indicating progressive convergence during training, and their close alignment suggests limited overfitting in the evaluated splits. Accuracy rises rapidly within the first ten epochs and then stabilizes around 97–98%, while validation accuracy remains close to training accuracy throughout most epochs. Recall also exceeds 90%, and precision remains above 95% near the end of training, indicating strong predictive performance within this dataset. These results support consistent learning behavior across the reported splits, although they should not be interpreted as conclusive evidence of broad generalization beyond the current experimental setting.

Model explainability

We analyze Grad-CAM visualizations across Dataverse, Tanzania, and Plagues Maiz and interpret the highlighted regions in terms of symptom morphology and center–boundary lesion structure. The heatmaps show that MaizeFormerX primarily bases decisions on disease-relevant tissue (lesion cores, chlorotic margins, streak continuity) rather than background clutter, indicating lesion-grounded reasoning. Importantly, this behavior is consistent with the role of the Cross-Scale Attention Fusion (CSAF) module, which integrates fine-

Augmentation	vs. MobileViT	vs. EfficientFormer	vs. TinyViT	vs. Swin Transformer V2
Dataverse				
×2	$p = 0.0025$	$p = 0.0047$	$p = 0.0069$	$p = 0.0092$
×4	$p = 0.0017$	$p = 0.0039$	$p = 0.0051$	$p = 0.0075$
×6	$p = 0.0031$	$p = 0.0042$	$p = 0.0059$	$p = 0.0088$
Tanzania				
×2	$p = 0.0022$	$p = 0.0040$	$p = 0.0064$	$p = 0.0086$
×4	$p = 0.0033$	$p = 0.0048$	$p = 0.0057$	$p = 0.0090$
×6	$p = 0.0019$	$p = 0.0037$	$p = 0.0052$	$p = 0.0078$
Plagues Maiz				
×2	$p = 0.0028$	$p = 0.0043$	$p = 0.0067$	$p = 0.0095$
×4	$p = 0.0016$	$p = 0.0035$	$p = 0.0050$	$p = 0.0082$
×6	$p = 0.0021$	$p = 0.0044$	$p = 0.0061$	$p = 0.0091$

Table 13. Statistical significance (p-values) comparing MaizeFormerX against other models across datasets and augmentation levels using the Wilcoxon signed-rank test.

Model	Params (M)	GFLOPs @224 × 224	CPU Latency (ms) (i9-13900 K)	GPU Latency (ms) (RTX 4090)
ShuffleNetV2 (1.0×)	2.2	0.14	7.4	0.68
GhostNet (1.0×)	5.5	0.16	6	0.74
MobileNetV3-Large	5.1	0.23	9.8	0.83
EfficientNet-B0	5.8	0.37	8.1	0.95
EfficientNet-B1	7.3	0.71	12.9	1.18
ShuffleNetV2 (1.5×)	3.7	0.31	11.2	0.93
GhostNet (1.3×)	7	0.28	9.3	1.02
MixMobileNet	4.4	0.21	12.1	0.97
MobileViT	5.9	0.7	14.7	1.22
EfficientFormer	12.6	1.15	13.3	1.35
TinyViT	10.4	2.12	21.8	1.78
Swin Transformer V2 (Tiny)	29.4	4.1	27.4	2.36
MaizeFormerX (Ours)	9.6	1.05	16.4	1.42

Table 14. Model complexity and inference efficiency comparison of MaizeFormerX with lightweight CNN and ViT baselines.

scale texture evidence (e.g., small necrotic speckles or streak granularity) with coarse-scale contextual cues (e.g., lesion spread, continuity, and global discoloration), producing saliency that is spatially coherent and symptom-consistent. To explicitly analyze CSAF's contribution to explainability, we further visualize (i) Grad-CAM at the CSAF output, (ii) CSAF cross-scale attention (CLS→coarse tokens), (iii) Grad-CAM for an ablated variant without CSAF, and (iv) the difference map ΔCAM = (full – no-CSAF) (Fig. 19). The attention overlays show that CSAF increases activation on lesion cores and chlorotic/necrotic margins while suppressing background-driven responses (e.g., soil texture, shadows, and specular highlights). Removing CSAF produces more diffuse saliency and weaker boundary localization, confirming that cross-scale fusion improves not only classification accuracy but also explanation fidelity. Dataverse (Fig. 20). The Grad-CAM outputs demonstrate class-specific localization patterns. For the Healthy class, the model assigns higher importance to the central midrib and major venation, reflecting an intact structural pattern with no symptomatic disruption. For MLB, activations concentrate along the elongated necrotic lesions and their surrounding margins, indicating that the model leverages both lesion cores and boundary transitions that separate affected from healthy tissue. For MSV, the heatmaps align with the striped chlorosis patterns, emphasizing streak continuity along the leaf blade an important cue for viral infection. These patterns indicate that MaizeFormerX is not responding to uniform color shifts alone; instead, it attends to discriminative spatial arrangements of symptoms, supporting interpretable predictions.

Tanzania (Fig. 21) shows similar trends in interpretability despite greater variability in appearance. For Healthy plants, attention is focused on the unblemished tissue with a stable venation structure, indicating resilience to differences in acquisition. In the case of MLN, the heatmaps highlight areas with a mosaic-like pattern of chlorosis and necrosis, showing that the model emphasizes both the affected regions and the surrounding boundaries that define symptomatic tissue. For MSV, the activations are more broadly distributed along streaks, capturing the overall spread of viral symptoms. Notably, compared to purely local explanations, these maps demonstrate multi-scale reasoning: fine-grained indicators, such as small irregularities in necrotic texture, are supported by coarser contextual elements, such as the extent and direction of streaking. This aligns with CSAF's objective of cross-resolution fusion.

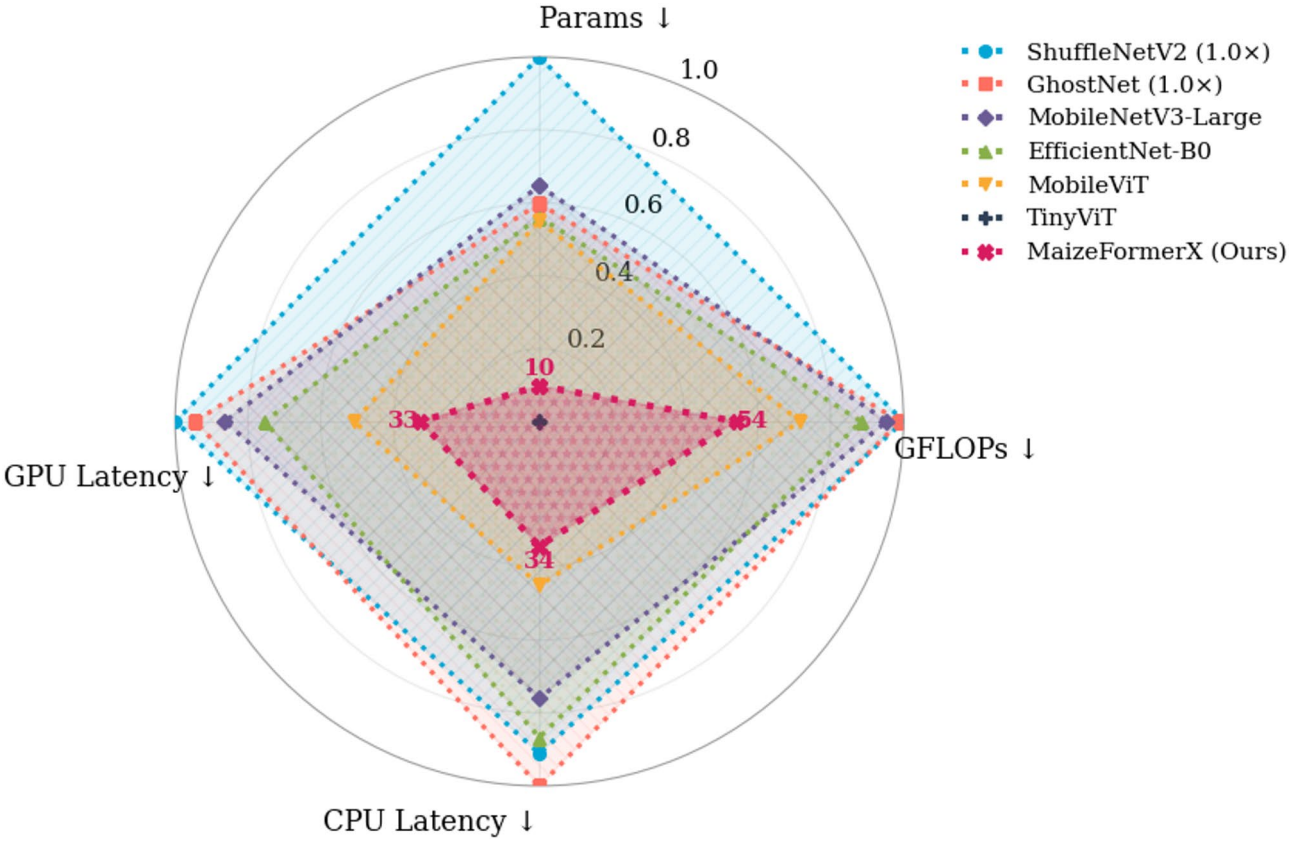


Fig. 15. Efficiency radar of MaizeFormerX vs. lightweight CNN/ViT baselines (224 × 224, batch = 1). Axes show inverse-normalized cost scores (0–100) for Params, GFLOPs, CPU latency (i9-13900 K), and GPU latency (RTX 4090); higher is better.

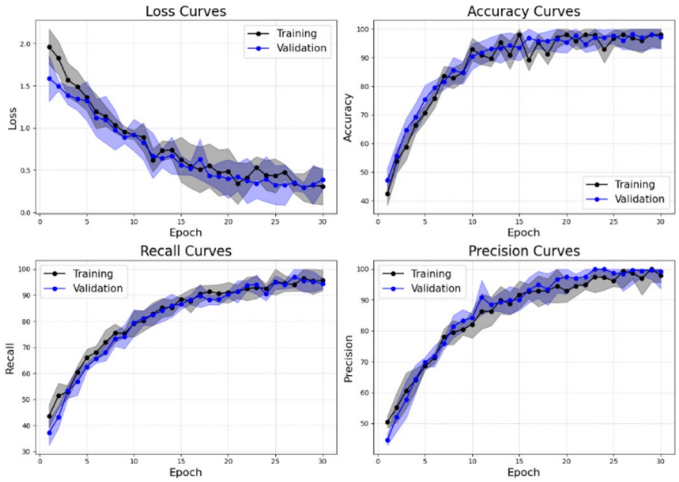


Fig. 16. Dataverse learning curve for MaizeFormerx.

The Plagues Maiz dataset (Fig. 22) incorporates a variety of bacterial, fungal, and viral classifications, which provides a robust evaluation of explanation quality. The Grad-CAM maps consistently align with symptoms across multiple pathogens. For bacterial examples, such as "Erwinia chrysanthemi" and "Erwinia stewartii", the model focuses on areas showing water-soaking or edge rot, particularly at the margins where tissue degradation sharply contrasts with healthy surrounding tissue. In the case of fungal infections, the activations often concentrate on localized clusters of lesions and distinct necrotic centers, while also capturing surrounding chlorotic transitions that indicate disease progression. For viral and mixed infections, the heatmaps highlight streak-like patterns and

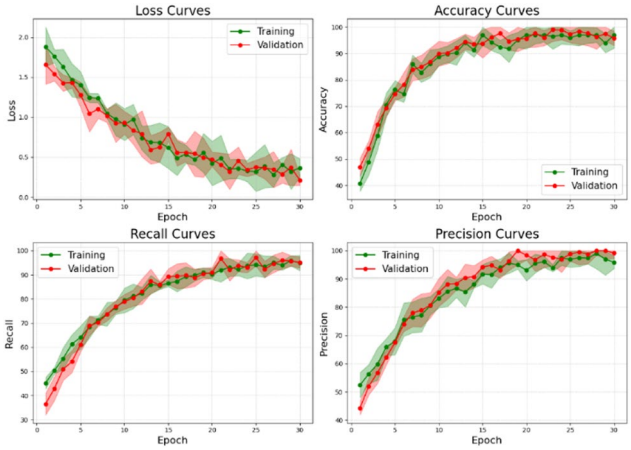


Fig. 17. Tanzania learning curve for MaizeFormerx.

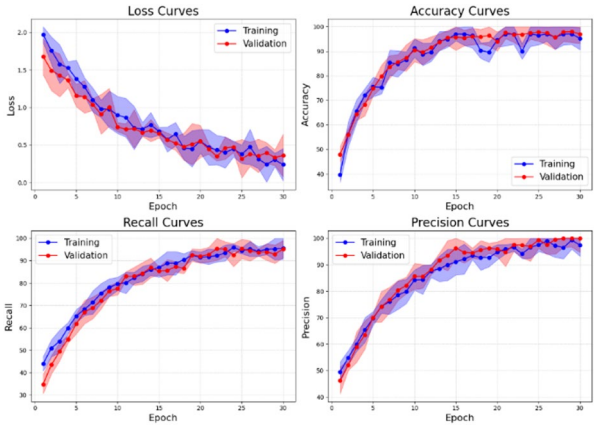


Fig. 18. Plagues Maiz learning curve for MaizeFormerx.

larger-scale discolorations instead of isolated background regions. This observation suggests that MaizeFormerX is learning about disease morphology rather than picking up on unrelated context.

In addition to that, we recommend reporting a lightweight quantitative sanity check (to be included as a small table) measuring how much saliency is concentrated on symptomatic tissue. A practical choice is the Lesion Focus Ratio, computed on a small manually annotated subset (e.g., loose lesion bounding boxes):

$$\text{Focus Ratio} = \frac{\sum \text{CAM inside lesion ROI}}{\sum \text{CAM on leaf region}}.$$

Higher values indicate more lesion-grounded explanations and reduce the risk that predictions rely on background artifacts.

Misclassified cases in Fig. 23 highlight failure modes tied to symptom overlap and noise. In the Dataverse, MLB (Mosaic Leaf Blight) and MSV (Mosaic Virus) can be confused when elongated lesions resemble streaky discoloration due to lighting variations, resulting in diffuse activations. Healthy samples misidentified as diseased often show scattered saliency in high-contrast areas, indicating sensitivity to lighting artifacts or minor texture changes. In Tanzania, the model occasionally misclassifies MSV as MLN (Mosaic Leaf Necrosis) when mosaic patterns and streaking are present or small necrotic patches are only weakly visible; this shifts heatmaps from localized lesions to broader discoloration. In Plagues Maiz, closely related bacterial classes show overlapping activations in water-soaked margins and texture patterns, while some fungal and viral samples display scattered attention when symptoms are subtle or mixed. These findings suggest that future improvements, such as lesion-guided region of interest (ROI) extraction and boundary-aware attention regularization, could enhance explanation fidelity.

Web application and explainability

The Maize Leaf Disease Monitor is a web-based prototype designed for the classification and visual interpretation of maize leaf conditions (see Fig. 24). It offers essential functions such as dataset selection, image upload, class

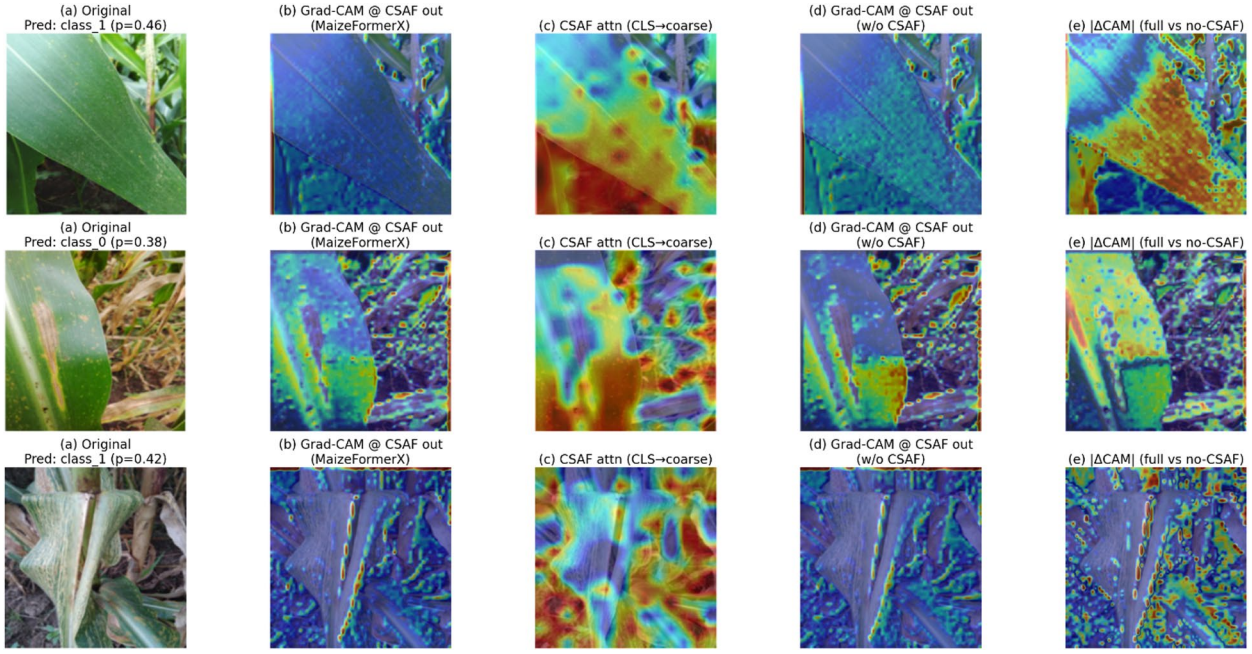


Fig. 19. CSAF-driven explainability analysis. For representative samples, we show (**a**) the original leaf image, (**b**) Grad-CAM computed at the CSAF output for MaizeFormerX, (**c**) CSAF cross-scale attention (CLS→coarse tokens; averaged over heads), (**d**) Grad-CAM at the same layer for the ablated model without CSAF, and (**e**) the difference map ΔCAM = (full – no-CSAF). CSAF increases lesion-centered and boundary-aware activations while reducing background-driven saliency.

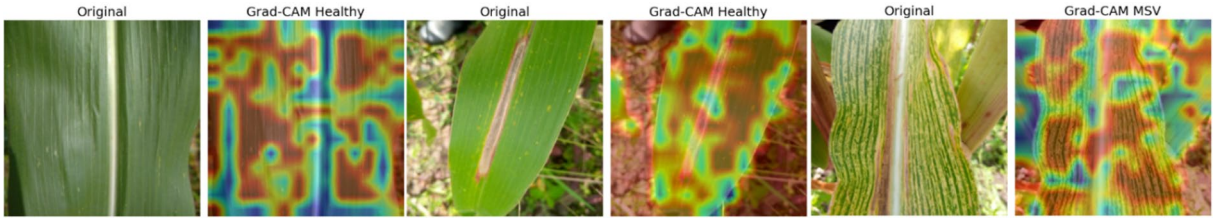


Fig. 20. Dataverse XAI outputs for original maize leaf images are shown alongside their corresponding Grad-CAM outputs, highlighting discriminative regions used by the model to classify affected samples.

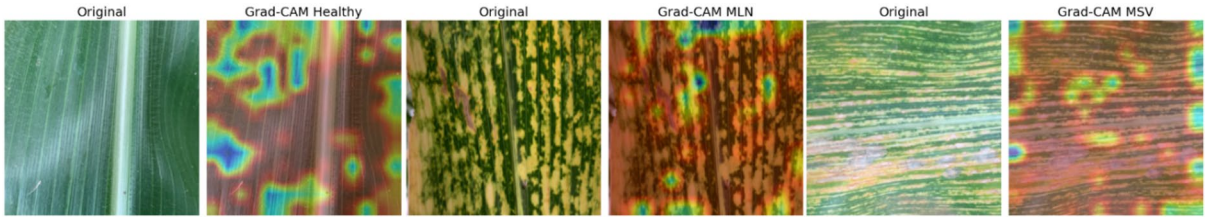


Fig. 21. Grad-CAM visualizations on the Tanzania dataset. The original leaf images are shown alongside their corresponding Grad-CAM heatmaps, highlighting key regions used by the model to identify conditions.

prediction, and Grad-CAM-based visualization, creating an accessible workflow for both technical and non-technical users. The system supports multiple datasets, including Dataverse, Tanzania, and Plagues Maiz, each associated with its corresponding pretrained model. Users can upload an image of a maize leaf, select the relevant dataset, and receive a predicted class, such as "Healthy," along with a Grad-CAM heatmap that highlights the regions of the image that influenced the model's decision. This integration of explainable AI enhances prediction transparency and allows for qualitative visual inspection of the model's attention.

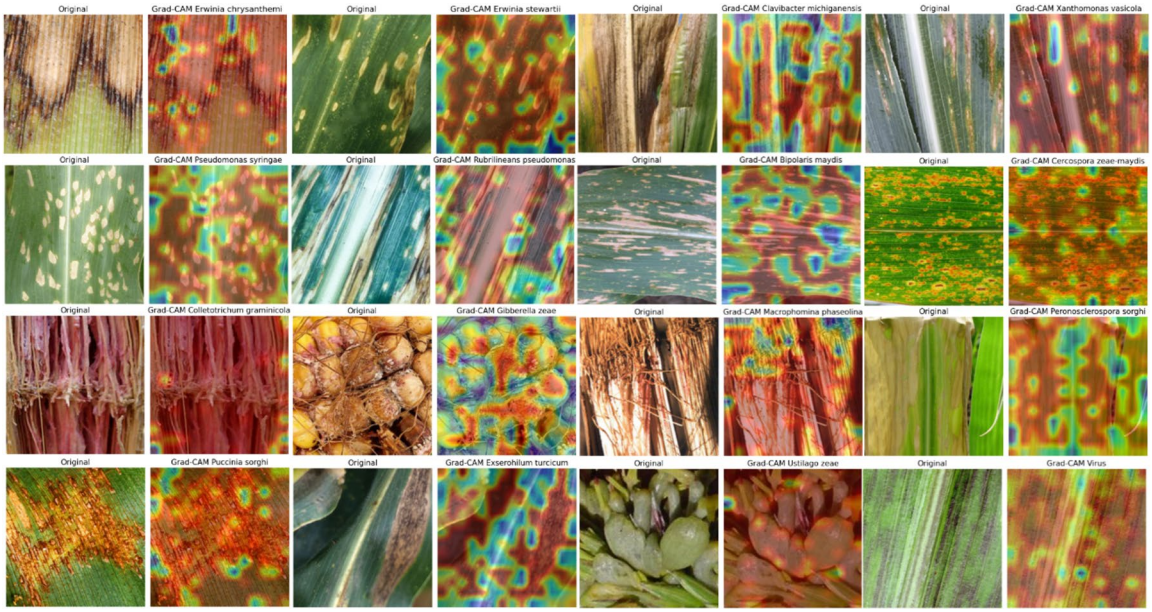


Fig. 22. Grad-CAM visualizations for the Plagues Maiz dataset. The original leaf images are paired with corresponding Grad-CAM outputs, showcasing the model's focus regions across various disease classes.

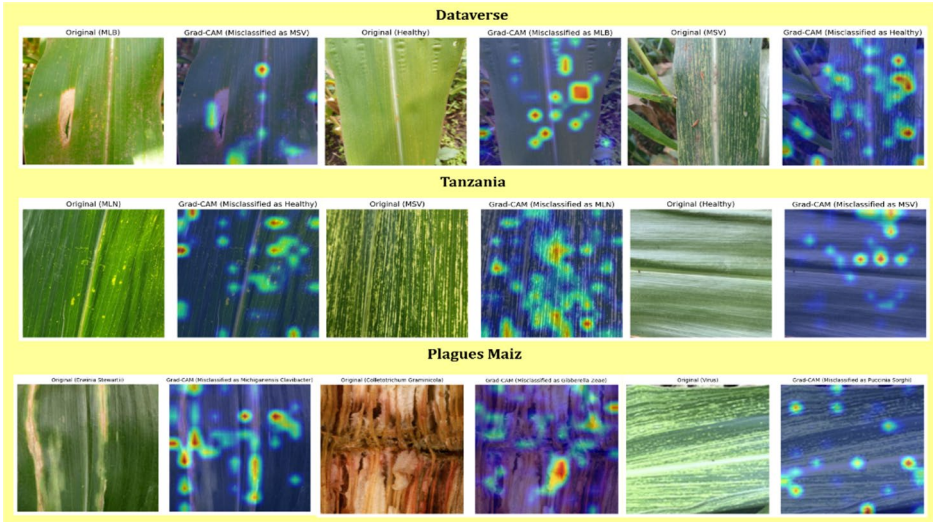


Fig. 23. Grad-CAM visualization of misclassified samples across all datasets. Each pair shows the original image and the corresponding heatmap highlighting attention regions contributing to misclassification.

The application features a modular backend that loads pretrained models based on the selected dataset and utilizes stateless API endpoints to minimize memory overhead and support parallel inference. Input images are resized to 224×224 pixels and preprocessed on the client side to improve efficiency. Grad-CAM visualizations are generated asynchronously to decouple saliency computation from the forward-pass latency. To enhance spatial detail, the application extracts intermediate representations from the lower layers of the Swin Transformer V2, although there are known limitations to Grad-CAM interpretation in hierarchical transformer architectures. At its current stage, the system should be viewed as a prototype interface for demonstration and controlled evaluation rather than a fully field-ready platform. Future developments may involve Docker-based deployment, on-device preprocessing using OpenCV and ONNX Runtime, model caching, INT8 quantization, and uncertainty estimation methods such as Monte Carlo Dropout and temperature scaling. These planned enhancements aim to improve robustness, decrease backend latency during model switching, and increase reliability in more challenging real-world conditions.

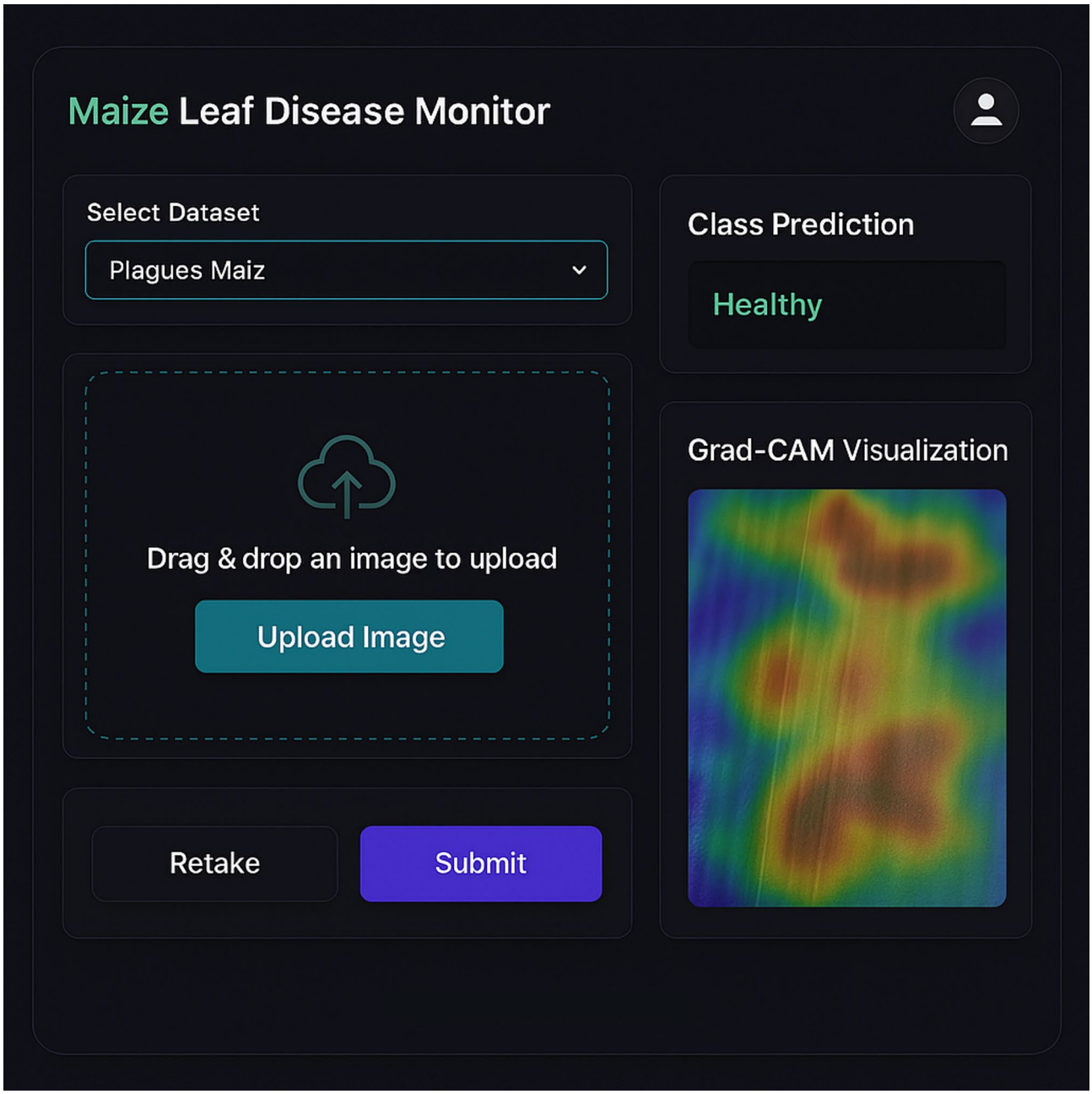


Fig. 24. Web interface of the Maize Leaf Disease Monitor. Users can upload maize leaf images, select a dataset, and receive class predictions and Grad-CAM visualizations for interpretable disease classification.

Comparison with the previous studies

As shown in Table 15, many studies have utilized CNN-based architectures, such as Jasrotia et al.[22], Elmasry et al.[24], and Masood et al.[25], achieving high accuracy (96–98%) on datasets like PlantVillage or small custom collections. Malik et al.[26] reported an accuracy of 99.8% using TreeNet; however, this result was achieved on clean, lab-based data, limiting applicability under field conditions characterized by noise and variable lighting. ViT and hybrid CNN–ViT models have also been explored, with Ramadan et al.[30] and Isinkaye et al.[32] achieving accuracies of 93–95%, and Gole et al.[35] reporting 96.93% with TrIncNet. However, these models often require significant computational resources and have not been rigorously tested across diverse, real-world environments, limiting their scalability for potential field deployment.

A critical gap in most existing works is the absence of XAI. While Elmasry et al.[24], Aboelenin et al.[38], and Ji et al.[43] incorporated Grad-CAM or similar methods, the majority of studies lack interpretability features essential for building trust and enabling actionable decision-making in agricultural contexts. Additionally, few studies have developed web-based applications; In contrast, MaizeFormerX employs a lightweight, multi-scale ViT architecture validated on large and diverse datasets, including Dataverse (75,936 images), Tanzania (44,916

Reference	Proposed Model	Dataset Name	No. of Data	Accuracy (%)	Web App	XAI
Jasrotia et al.[22]	CNN + CLAHE-HSV	PlantVillage	3,852	96.76	No	No
Elmasry et al.[24]	DenseNetDNN	Custom (4-class)	4,188	96.1	No	Yes
Masood et al.[25]	MaizeNet	Field Images	2,112	97.89	No	No
Malik et al.[26]	TreeNet	PlantVillage	4,000	99.8	No	No
Thakur et al.[27]	VGG-ICNN	Custom Maize Subset	481	91.36	No	No
Rajeena et al.[28]	EfficientNet B0	PlantVillage + PlantDoc	3,188	98.85	No	No
Ramadan et al.[30]	ViT-B/16	Custom (4-class)	4,188	94.51	No	No
Ramadan et al.[31]	CycleGAN + DenseNet169	Custom (2-class)	4,600	98.48	No	No
Isinkaye et al.[32]	VAE + ViT	Corn/Potato/Tomato	27,872	93.2	No	No
Mayo et al.[34]	ViT (Custom)	Tanzania (MLN/MSV)	26,913	93.1	No	No
Gole et al.[35]	TrIncNet (ViT + Inception)	PV + Maize Field	13,971	96.93	No	No
Li et al.[36]	Sim-ConvNeXt	Custom Maize	17,670	95.2	No	No
Zhu et al.[37]	State-Space Attention DL	Custom Maize	7,343	94	No	No
Aboelenin et al.[38]	Hybrid CNN–ViT	PlantVillage (Corn)	6,774	98	No	Yes
Wang et al.[42]	ULEN	PlantVillage (Maize)	3,852	97.76	No	No
Ji et al.[43]	ICS-ResNet	PlantVillage + Kaggle	8,280	98.87	No	Yes
Ours	MaizeFormerX	Dataverse	75,936	97.8	Yes	Yes
		Tanzania	44,916	97.5	Yes	Yes
		Plagues Maiz	4,080	96.9	Yes	Yes

Table 15. Comparison of existing studies on maize leaf disease classification

images), and Plagues Maiz (4,080 images). It achieves high accuracy (96.9–97.8%). It combines interpretability with a web-based prototype interface, positioning it as a research framework oriented toward deployment rather than being just a laboratory model. MaizeFormerX is a promising candidate for scalable maize disease monitoring across various agroecological settings. However, its readiness for practical field use and operational deployment still needs validation through real-world trials, user-centered evaluations, and on-device testing.

Discussion

This study proposed a lightweight and explainable ViT designed for real-world maize disease classification. By integrating a three-level multi-scale embedding and a CSAF module, this model achieves high accuracy while maintaining computational efficiency. Its web-based XAI platform provides interpretable lesion heatmaps. The scale-adaptive transformer encoder of MaizeFormerX effectively preserves lesion texture and spatial context across various field conditions. It outperformed larger ViT models, such as Swin V2 and TinyViT, in both accuracy and efficiency. In a broader transformer context, GeNetFormer[52] shows how transformer representations can be applied in complex real-world scenarios, but its focus and data type differ from the current study, making it unsuitable for direct comparison with maize leaf disease identification. In contrast, MaizeFormerX is specifically built for recognizing agricultural symptoms. It features cross-scale attention fusion to integrate detailed lesion textures with broader contextual information, while ensuring efficiency in terms of parameters, FLOPs, and latency. Furthermore, the pipeline incorporates interpretability tools like Grad-CAM and CSAF attention and tests its robustness across various acquisition domains. On the Tanzania dataset, optimal performance was reached at 2× augmentation, as further augmentation resulted in slight overfitting, which reduced the MCC by 1–2%. For the imbalanced Plagues Maiz dataset, the best performance occurred at 4× augmentation, with 6× causing degradation in the recognition of minority classes due to distribution shifts from aggressive augmentation.

The model exhibited strong cross-dataset generalization, achieving 86.8% accuracy when trained on the Tanzania dataset and tested on the Plagues Maiz dataset, surpassing baseline models by 3–6%. However, the benefits of augmentation plateaued beyond 2×. Error analysis indicated significant confusions, particularly between Healthy and MSV, as well as MLN and MSV, caused by visually similar patterns like chlorotic streaks. Moderate augmentation (4×) helped reduce these errors, while excessive augmentation distorted key features. Techniques such as rotation, histogram equalization, and color jitter improved the MCC by up to 4.8%, illustrating the importance of dataset-specific augmentation strategies.

Explainability results demonstrated clear attention to critical regions: midribs in healthy leaves, elongated lesions in MLB, and mosaic patterns in MSV/MLN. Diffuse saliency in misclassified cases indicates that future enhancements could involve boundary-aware attention or lesion-focused cascaded classification. Ablation studies confirmed the importance of the CSAF module; removing it decreased both accuracy and MCC by 2–4%. The transformer encoder stack contributed an additional 1–2%, while multi-scale fusion provided a further 1% gain, particularly valuable for datasets with high inter-class similarity. Statistical validation (p-values < 0.01) across all augmentation levels supports the model's robustness, with optimal alignment occurring at 4× augmentation. Compared to previous CNN-based approaches, which often lack real-world validation and interpretability, MaizeFormerX strikes a balance between performance and practicality.

The practical relevance of MaizeFormerX is demonstrated through a web application integrated with Grad-CAM, which supports early-stage disease screening and enhances transparency by allowing for visual inspection of model predictions. Due to its relatively lightweight design, this model could be adapted for use on resource-constrained platforms, such as UAV-assisted systems, IoT-based agricultural devices, and mobile applications, although these deployment scenarios were not experimentally tested in this study. Additionally, its computational profile suggests the potential for more efficient training compared to significantly larger ViT-based models, though no formal energy consumption benchmarks were conducted. Furthermore, its compatibility with REST-based interfaces indicates the possibility of integrating it into broader precision agriculture pipelines, including sensor- and drone-assisted monitoring systems, which is an important direction for future implementation work.

Despite its strengths, several limitations exist. First, using only RGB imagery limits recognition of early-stage or subtle symptoms that might be better identified with multispectral, hyperspectral, or thermal imaging. Second, while MaizeFormerX is lighter than larger models, it still faces challenges when deployed on ultra-low-power edge devices without extra compression, memory optimization, and thermal control. Third, class imbalance and regional cultivar diversity can skew results toward majority classes; although augmentation helps, excessive use can lead to feature drift and distribution issues. Fourth, differences in imaging conditions (like lighting and device quality) can affect cross-region performance. Finally, Grad-CAM explanations can be vague, especially for mixed or subtle symptoms, which complicate interpretation without precise validation.

Future work will focus on addressing these limitations by (i) integrating RGB with multispectral/hyperspectral/thermal inputs for better symptom classification, (ii) enhancing robustness through adversarial domain adaptation and domain-invariant augmentation to minimize drift, and (iii) optimizing efficiency with quantization-aware training and structured pruning to support low-power devices like ESP32 and Cortex-M. To combat class imbalance, we will implement class-balanced sampling, focal loss, and other strategies, and explore GAN/VAE-based methods for enhancing rare classes. For interpretability, we will complement Grad-CAM with Layer-wise Relevance Propagation (LRP), SHAP, and additional metrics for better insight. Moreover, modern real-time detection models like Ultralytics YOLO26 offer a valuable approach for future experiments in lesion localization and practical deployment. YOLO26's efficient detection strategies are particularly suited for agricultural imaging, where lesions can be tiny. One promising strategy is a two-stage pipeline where a detector identifies potential lesion areas, and MaizeFormerX classifies them. This could enhance robustness and provide clearer explanations. Lastly, we aim to improve scalability through drone capture, real-time sensor integration, and collaboration with digital farm platforms, followed by field trials to assess usability and decision support in real-world agriculture.

Conclusion

This study presents an efficient and scalable solution for image-based classification of maize leaf diseases in precision agriculture. Across the evaluated datasets, the model achieved stronger MCC and PR-AUC performance than the ViT baselines considered in this study. The cross-dataset results further suggest encouraging transferability under the shared-label evaluation setting, although these findings should not be interpreted as conclusive evidence of full domain invariance or universal generalization without additional validation. The proposed Grad-CAM-enabled web application also demonstrates how model predictions can be accompanied by visual explanations, allowing agronomists and other users to upload leaf images and inspect salient regions for qualitative assessment and potential decision-support use. There are several limitations to address. Class imbalance in rare disease categories and residual bias in the dataset may impact performance for underrepresented cases. Additionally, current XAI outputs may lack clarity for complex visual samples. To enhance the evaluation of MaizeFormerX, broader benchmarking under diverse environmental and field conditions is essential. Future work will focus on improving temporal and contextual understanding through spatiotemporal modeling, integrating multimodal data, and enhancing domain adaptation for consistent performance across varied conditions. We will also explore model compression and optimization for deployment on resource-constrained devices, along with federated learning strategies for privacy-aware updates. These efforts aim to transition MaizeFormerX from an experimental framework to a practical tool for maize health monitoring and agricultural decision support.

Data availability

The datasets used in this study are publicly available and sourced from Dataverse (https://doi.org/10.7910/DVN/LPGHKK), Tanzania (https://doi.org/10.17632/fkw49mz3xs.1) and Plagues Maiz (https://figshare.com/articles/MaizePD/10314539/3) . All code, preprocessing pipelines, and experimental configurations used in this work are available at: https://github.com/rezaul-h/MaizeFormerX/.

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Declarations

Conflict of interest

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Additional information

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